

Research Report on

Mechanism-oriented comparisons of time-series clustering similarity measures across noise, misalignment, and length

Full search query: I want to find prior work, theoretical foundations, and experimental design literature for a mechanism-oriented study of general-purpose time-series clustering that systematically compares cross-paradigm similarity measures—lock-step (ED), elastic (DTW, MSM), sliding/alignment-based (SBD), and distributional kernel methods (especially IDK)—and examines whether this comparison is novel and how noise level, temporal misalignment, and sequence length shape their relative clustering performance, strengths, and weaknesses

Summary

There is no existing study that jointly performs a *mechanism-oriented*, general-purpose clustering comparison across **lock-step (ED)**, **elastic (DTW/MSM)**, **sliding/SBD**, and **distributional/kernel (IDK-like)** distances while systematically varying **noise**, **temporal misalignment**, and **sequence length**; the closest components are spread across benchmarking works on distances and clustering [2,3,4,5,7,8], a robustness-oriented domain study [1], invariance/stability validation [9], and property-based distance characterizations [4,10] plus an initial distributional-clustering approach [6].

Overall State of the Field Relative to Your Goal

What Exists vs. What You Want

- **What you want:** a *general-purpose, mechanism-oriented* study of time-series clustering that:
 - Compares **ED**, **elastic distances** (especially DTW, MSM), **sliding/SBD-like**, and **distributional/kernel (IDK-type)** similarity.
 - Uses **clustering** (not just classification) as the main task.
 - **Systematically manipulates:**
 - Noise level,
 - Temporal misalignment (global phase, local warping),
 - Sequence length,
 - And interprets results in terms of **mechanisms/inductive biases** of each paradigm.
- **What exists** (within [1,2,3,4,5,6,7,8,9,10]):
 - Large, rigorous **distance** and **clustering** benchmarks spanning multiple paradigms (ED, DTW, SBD, some kernels), but with *minimal explicit control* over noise, misalignment, and length [2,3,4,5,7,8].
 - A **domain-specific robustness study** that varies dataset properties (noise, outliers, cluster balance) and compares ED, multiple elastic measures (DTW, MSM, TWED), and SBD-like distances in clustering [1].
 - **Theoretical/empirical distance taxonomies** and **property-based analyses** that characterize invariances and noise sensitivity across many measures [4,10].
 - An **invariance-guided stability** framework for clustering validation that explicitly uses shift/warp/noise perturbations [9].
 - A small but relevant **distributional/distance** clustering approach using ED, DTW, KL, and Wasserstein on distributions of differences [6].

Key conclusion for novelty: integrating *all four similarity paradigms* into a single clustering study with **factorial control** over noise, misalignment, and length—while aligning clustering objectives to each distance's geometry—would be genuinely new relative to current work.

Cross Paradigm Similarity: What Has Been Compared Already?

Benchmarking Distances and Paradigms (Mainly for Classification or Generic Performance)

- Paparrizos et al. 2020 – “Debunking Four Misconceptions” [4]
 - **Scope:** 71 time-series distances on 128 datasets, chiefly for classification.
 - **Coverage:**
 - 52 **lock-step**, 4 **sliding**, 7 **elastic**, 4 **kernel**, 4 **embedding** distances.

- **Contributions:**
 - Establishes a **taxonomy** spanning the paradigms you care about.
 - Shows that:
 - DTW is **not universally superior** to ED.
 - Normalization choices dramatically affect performance.
 - Some non-DTW distances (including sliding and kernel ones) can be highly competitive.
- **Limitations for your goal:**
 - Focus is **classification**, not clustering.
 - No explicit manipulation of noise, misalignment, or length.
 - Kernels are present but not coupled to **kernel-based clustering**.
- **d'Hondt et al. 2025 – Multivariate Distance Measures [3]**
 - **Scope:** 30 multivariate distances across 8 categories (including lock-step, elastic, sliding, kernel, embedding), 13 normalization methods, 2 channel-dependency models, 3 tasks (incl. clustering).
 - **Relevance:**
 - Extends [4]'s distance taxonomy to **multivariate** series and includes **kernel** and **sliding** measures in a unified evaluation.
 - Confirms that many univariate findings extend to multivariate distances.
 - **Limitations:**
 - Task mix, not clustering-only; no explicit noise/misalignment/length sweeps.

Use for your work: [4] and [3] are the core **distance-taxonomy and performance-baseline** references; they validate which measures are worth including from each paradigm and emphasize the critical role of normalization.

Cross Paradigm Clustering Comparisons (ED vs DTW vs SBD)

- **Javed et al. 2020 – Benchmark Study on Time Series Clustering [2]**
 - **Scope:** 8 clustering algorithms × 3 distances (ED, DTW with fixed 5% Sakoe–Chiba window, SBD) on 112 UCR datasets.
 - **Findings:**
 - ED is often competitive in clustering; DTW does not dominate.
 - SBD (shape-based, shift/scale-invariant) is typically competitive with or superior to DTW in many settings.
 - **Strengths:**
 - **First large, systematic clustering benchmark** across **lock-step, elastic, and sliding** measures.
 - Strong control of **preprocessing** (equal length, uniform normalization).
 - **Limitations:**
 - No kernel or distributional methods.
 - Dataset noise and misalignment are not experimental factors; they're whatever UCR happens to contain.
 - No explicit analysis of **sequence length** effects beyond the archive's diversity.
- **Holder & Bagnall 2022 – Elastic Distance Functions for Clustering [5]**
 - **Scope:** 9 elastic distances (DTW, DDTW, WDTW, ERP, EDR, LCSS, TWE, MSM, WDDTW) plus ED on UCR datasets; k-means, k-medoids, and DTW Barycenter Averaging (DBA).
 - **Key mechanistic insights:**
 - With **k-means**, DTW often performs **worse than ED** because Euclidean means are poor prototypes in DTW geometry.
 - With **k-medoids**, *all* elastic distances improve; **MSM** and **TWE** often outperform DTW.
 - Distances that combine editing + warping (MSM, TWE) tend to outperform pure DTW: they are more **metric-like** and less prone to over-warp.
 - **Limitations:**
 - Primarily an **elastic vs ED** bake-off; no SBD or distributional/kernels.
 - No explicit perturbation of noise or misalignment; equal-length, z-normalized UCR.

Use for your work: [2] gives you **ED vs DTW vs SBD** clustering baselines; [5] gives deep, **mechanistic insight into elastic distances vs clustering objective**, crucial for designing appropriate prototypes (means/medoids/barycenters) in your study.

Broad Clustering Benchmarks (Many Methods, Distances Implicit)

- **Odyssey and subsequent clustering benchmark:**
 - **Odyssey engine [8]:** an infrastructure for evaluating 80 clustering methods across 9 classes on 128 datasets, enabling consistent preprocessing and parameter sweeps.
 - **Paparrizos & Teja 2025 [7]:** builds on Odyssey to compare **84 clustering methods** across **10 classes** (including deep learning and foundation models).
 - Shows that **elastic measures are not uniformly superior**, and **deep learning does not automatically win**.
- **Relevance:**
 - Provide the **state-of-the-art empirical landscape** of clustering methods under standardized conditions.
 - Show that **distance choice and its interplay with algorithm class** remain critical.
- **Limitations:**
 - Not distance-centric; they treat distances as one component among many.
 - No deliberate manipulation of noise, misalignment, or length.

Use for your work: these are the **global clustering baselines** your more focused, mechanistic study can be positioned against; you can also reuse Odyssey-style infrastructure.

Noise, Misalignment, and Length: How They've Been (Partially) Studied

Explicit Perturbation / Robustness Studies

- **Smart-meter clustering with controlled dataset properties – Yerbury et al. 2024 [1]**
 - **Context:** clustering smart-meter load profiles.
 - **Distances:** ED variants, multiple **elastic** distances (DTW, MSM, TWED, etc.), and **sliding/k-sliding (SBD-like)** among others.
 - **Experimental design:**
 - Phase 1: assess 31 distances and 8 representations via leave-one-out classification.
 - Phase 2: select top distances and evaluate 11 clustering algorithms.
 - Phase 3: **robustness experiments** using **expert-informed synthetic datasets** that systematically vary:
 - **Noise levels,**
 - **Outlier prevalence,**
 - **Cluster balance.**
 - Synthetic data encode realistic **peak consumption patterns** with temporal variability (implicitly handling local shifts/warping).
- **Key findings:**
 - Distances that **allow local temporal shifts while preserving amplitude** (especially DTW and k-sliding/SBD-like) consistently outperform simple lock-step measures on this domain.
 - Constrained DTW (windowing) improves performance and efficiency and appears more robust.
 - Hierarchical clustering with Ward's linkage combined with these distances shows strong robustness across varied dataset properties—even with randomly sampled parameters.
- **Limitations:**
 - Misalignment is present in the generative process but not **factorized as a scalar parameter** (e.g., warp amplitude).
 - **No distributional/kernel** distances.
 - Sequence length is fixed by domain (daily/weekly profiles), not varied.

Use for your work: [1] is the most **direct precedent** for a *robustness-oriented, cross-paradigm clustering study*; it demonstrates how to use **semi-synthetic data** to vary noise & dataset composition and shows that **elastic and sliding measures respond differently to timing variability**.

- **Invariance-guided stability criterion – Forest et al. 2021 [9]**
 - **Goal:** model selection for time-series clustering via **stability under perturbations**.
 - **Core idea:**
 - Encode known **invariances** (e.g. shift, warp, or noise robustness) as **perturbation operators**.
 - Assess clustering stability under these perturbations; select models (k, algorithm, distance) that are stable.

- **Mechanistic relevance:**
 - Provides a principled way to:
 - Design **warp/shift/noise perturbations** aligned with distance properties.
 - Quantify robustness to such perturbations, independent of the specific distance.
- **Limitations:**
 - Distance- and algorithm-agnostic; does not itself compare paradigms.

Use for your work: [9] is a **template for experimental design and validation**—you can adopt their invariance-aligned perturbations to probe how each similarity paradigm behaves under controlled noise and misalignment.

Property-Based Analyses of Noise & Misalignment (Outside Clustering)

- **Dove et al. 2023 – Distance measures in ecology [10]**
 - **Scope:** 42 distances evaluated on **16 properties** important for ecological time-series tasks.
 - **Properties include:**
 - **Sensitivity/invariance to:**
 - Additive noise,
 - Global shifts in time,
 - Local temporal warping,
 - Amplitude scaling and offset,
 - Outliers, etc.
 - **Outputs:**
 - A **property table** indicating which measures are robust to which transformations.
 - A selection procedure to choose distances given desired invariances.
 - **Relevance:**
 - Operationalizes many of the *theoretical expectations* about ED vs DTW vs SBD vs others in terms of **concrete transformation tests**.
 - Strong conceptual match to your interest in **mechanisms** (noise, misalignment).
- **Paparrizos et al. 2020 – Misconceptions [4]**
 - Although classification-oriented, [4] also:
 - Explicitly examines **normalization effects** across distances.
 - Reveals that many assumptions about robustness and ordering of measures (e.g., “DTW always best”) are false once you systematically vary preprocessing.
 - Includes a small set of **kernel distances**, showing they are **not universally superior**.

Use for your work: [10] and [4] effectively give you **empirical “property maps”** of many distances with respect to noise, phase shifts, and warping, which you can:

- Use to choose representative measures from each paradigm.
- Formulate **hypotheses** about their behavior under controlled perturbations in clustering.

Sequence Length: A Major Gap

- None of the papers [1,2,3,4,5,6,7,8,9,10] performs a **systematic, controlled study of sequence length** as an independent experimental factor in clustering:
 - Benchmarks use datasets with varying lengths, but length is never swept while holding the generating structure fixed.
 - Comments on length mostly concern **computational complexity** (DTW’s cost, FFT for SBD, etc.), not **clustering performance** as function of length.
- This is a **clear open gap** and one of the most straightforward axes for novelty in your work.

Distributional / Kernel / IDK Type Methods: Existing Foundations

Kernel and Embedding Measures in Benchmark Studies

- **Paparrizos et al. 2020 [4]** and **d’Hondt et al. 2025 [3]**:

- Include **kernel functions** and **embedding measures** as distinct categories alongside lock-step, sliding, and elastic distances.
- Show that:
 - Kernels can be competitive in classification/multivariate tasks, but do **not** dominate.
 - Their performance is highly **sensitive to parameterization and normalization**.
- These kernels are typically designed to capture **global similarity structures** (e.g., shape, autocorrelation) rather than explicit alignment.

Implication: kernel methods are present in the distance zoo, but their **role in clustering**, especially with **kernel k-means or spectral clustering**, is barely explored in a controlled, cross-paradigm way.

Distributional Distances in Clustering

- **Thanakulkairid et al. 2022 [6]**
 - Propose representing each time series by the **distribution of its first-order differences**, then comparing these distributions using:
 - ED, DTW, **KL divergence**, **Wasserstein distance**.
 - Perform clustering via **graph partitioning** on the resulting distance graph.
 - **Findings:**
 - ED and KL on these difference distributions outperform DTW and Wasserstein in their experiments.
 - **Relevance:**
 - This is one of the few papers where **distributional distances** (KL, Wasserstein) are used for **time-series clustering** in a general-purpose way.
 - Conceptually close to **IDK-like views** where a time series is an empirical distribution in some feature space.

Implication: there is an emerging but **very thin** base of work that treats time series as **distributions** in clustering; **kernelized variants** (e.g. MMD/IDK kernels, kernel mean embeddings) are not yet evaluated in cross-paradigm clustering or under controlled perturbations.

Experimental and Methodological Lessons for a Mechanism-Oriented Study

1. Control Preprocessing and Parameterization

- **From [2,3,4,7,8]:**
 - Use **consistent preprocessing** across distances (e.g., z-normalization vs alternative normalizations) and consider **normalization as an experimental factor**.
 - Implement **rigorous parameter search or clear parameter-setting principles**:
 - DTW window widths (e.g., as % of length; see [1,2]).
 - MSM/TWE operation costs [5].
 - SBD's normalization and correlation parameters [2].
 - Kernel bandwidths and decay parameters [3,4].
- **Practical implication:** your study should state explicit *parameter-calibration rules* that are comparable across paradigms (e.g., matching expected distance scales or regularization strength), so that observed differences can be attributed to **inductive bias**, not tuning.

2. Align Clustering Objective with Distance Geometry

- **DTW/elastic vs ED insights from [5]:**
 - Use **k-means with ED** (Euclidean centroids).
 - Use **k-medoids** or specialized barycenters for **DTW and MSM**.
 - For SBD, use **shape-based centroids** (k-Shape-style).
 - For kernels, use **kernel k-means** or spectral clustering, where centroids are mean embeddings.

Mechanistic value: when the geometry and prototype definition are aligned, performance differences reflect **distance properties** (e.g., warping invariance, distributional robustness) rather than artifacts of mis-specified centroids.

3. Design Factorial Synthetic Experiments

Building on [1,9,10]:

- **Data-generating factors:**
 - **Noise:**
 - Additive Gaussian / colored noise with controllable SNR.
 - Heavy-tailed or bursty noise to test robustness vs outliers.
 - **Misalignment:**
 - **Global shifts** (phase) to favor SBD-like methods.
 - **Local warping** (variable speed, local delays) to favor DTW/MSM.
 - Possibly combinations of both.
 - **Sequence length:**
 - Generate sequences from the same underlying pattern/process at different lengths L (e.g., via replication, sub-sampling, or extending the process).
 - Study how each distance's **discriminative power and stability** evolve with L .
- **Link to theoretical expectations:**
 - ED: sensitive to even small phase shifts; may “average out” noise on long sequences but suffer from high-dimensional distance concentration.
 - DTW/MSM: should benefit from local warping in moderate misalignment; risk over-warping noise [5].
 - SBD: should excel under pure global phase shifts but fail under local speed changes.
 - Distributional/IDK: should become more stable as sequences lengthen (better distribution estimates), and be robust to re-orderings and moderate misalignment, but sensitive to distributional distortions.
- **Outcome measures:**
 - Cluster recovery metrics (e.g., ARI, NMI) on synthetic data with **known generative clusters**.
 - **Stability measures** à la [9], with perturbations matched to hypothesized invariances.
 - Distance-property diagnostics inspired by [10] (e.g., measure of how cluster assignments change under controlled phase shifts).

4. Draw on Property Maps to Frame Hypotheses

- Use [10]’s **property-based catalogue** and [4]’s **taxonomy** to:
 - Choose a manageable subset of distances from each paradigm that span **different property combinations** (e.g., ED vs z-normalized Euclidean vs correlation-based; DTW vs WDTW vs MSM; pure SBD vs variants; a couple of distributional kernels such as MMD-like or IDK-style).
 - Form clear **mechanistic hypotheses**:
 - “As global phase variance increases (with no local warp), SBD methods should dominate ED and DTW.”
 - “As local warp amplitude increases (with minimal phase shift), MSM and TWE should outperform DTW with rigid windows and SBD.”
 - “As length increases with stationary noise, distributional kernels should gain relative advantage.”

Bottom Line for Your Planned Study

- **No existing paper** unifies:
 - **All four similarity paradigms** (ED, DTW/MSM, SBD, distributional/IDK kernels),
 - Under **clustering**, with
 - **Orthogonal, controlled sweeps** of **noise level**, **temporal misalignment**, and **sequence length**.
- **Closest building blocks**:
 - Broad distance taxonomies and performance maps: [4], [3].
 - Cross-paradigm clustering benchmarks (ED vs DTW vs SBD): [2], and elastic-centric but mechanistically rich: [5].
 - Robustness-oriented clustering under varying dataset properties (incl. noise): [1].
 - Invariance-guided perturbation and stability validation: [9].
 - Property-based characterization of noise and misalignment sensitivity: [10].
 - Early distributional/distance clustering: [6].
- **Your opportunity** is to **synthesize these strands** into a mechanism-oriented experimental framework that:

- Uses **well-chosen representatives** of each paradigm grounded in [3,4,10].
- Aligns the clustering objective to each distance as guided by [5].
- Employs **semi-synthetic data with factorized control** over noise, misalignment, and length inspired by [1,9].
- Treats **distributional/IDK kernels** as first-class comparators alongside ED, DTW/MSM, and SBD, thereby filling a clear gap in the literature.

Categories

Comparative Overview of the Most Relevant Papers

High Level Positioning Relative to a Mechanism Oriented, Cross Paradigm Study

The table below summarizes how each paper aligns with your target study: *general-purpose time-series clustering, cross-paradigm similarity (ED, DTW/MSM, SBD, distributional/kernels/IDK), and explicit analysis of noise, misalignment, and sequence length.*

Ref	Primary Task / Domain	Similarity Paradigms Covered	Focus on Clustering?	Treatment of Noise / Misalignment / Length	Role for Our Purposes
[1] Yerbury et al. 2024	Smart-meter load profiles (application-specific)	Lock-step (ED variants), elastic (DTW, MSM, TWED, etc.), sliding/SBD-like (k-sliding) , other specialized measures	Yes, systematic clustering evaluation (31 distances, 8 representations, 11 clustering algorithms)	Noise and outliers explicitly varied on synthetic data; local temporal shifts/warping via expert-influenced generative design; no explicit sequence-length study	Closest to a mechanism-aware robustness study in a real application domain , with cross-paradigm distance comparison but <i>no distributional/kernels</i> and only implicit misalignment modeling
[2] Javed et al. 2020	General-purpose (UCR archive, 112 datasets)	Lock-step (ED) , elastic (DTW) , sliding/SBD	Yes (8 clustering algorithms across distance types)	Carefully controlled <i>preprocessing</i> and <i>length (equal-length UCR)</i> , but no explicit noise/misalignment/length manipulation	Foundational cross-paradigm clustering benchmark (ED vs DTW vs SBD) but performance oriented, not mechanism oriented
[3] d'Hondt et al. 2025	Multivariate time series across several tasks	8 categories including lock-step, sliding, elastic, kernel, embedding measures (30 distances)	Partly (3 downstream tasks: classification, clustering, others)	No explicit systematic variation of noise/misalignment/length; some <i>normalization</i> and multivariate dependence analysis	Most comprehensive cross-paradigm distance study in multivariate setting , includes some kernel measures , but not mechanism-focused
[4] Paparrizos et al. 2020	General-purpose (128 datasets)	52 lock-step, 4 sliding, 7 elastic, 4 kernel, 4 embedding measures	Primarily classification; clustering not central	No designed manipulations of noise/misalignment/length; focus on misconceptions, normalization, and statistical tests	Core theoretical/empirical foundation on distance properties across paradigms , including kernel measures, but not clustering or mechanistic perturbations
[5] Holder et al. 2022	General-purpose (UCR univariate)	Elastic distances (DTW, MSM, ERP, TWE, etc.) + ED baseline	Yes (k-means, k-medoids, DBA centroids)	No explicit noise/misalignment/length manipulation; all equal-length & z-normalized	Deep dive into elastic vs Euclidean for clustering , with strong findings on MSM vs DTW, but <i>single paradigm family (elastic)</i>
[6] Thanakulkairid et al. 2022	General-purpose clustering via "distribution of differences"	ED, DTW , KL divergence , Wasserstein distance on distributions of first differences	Yes (graph partitioning clustering on distance graphs)	No systematic perturbation; implicit robustness via differencing and distributional comparison	Prototype of distributional / divergence-based approach applied to clustering, but no cross-paradigm mechanistic study
		Many method classes; underlying distances		Focus on benchmarking breadth & statistical comparisons; no controlled perturbation	State-of-the-art global benchmark for clustering methods , challenging the value

[7] Paparrizos & Teja 2025	General-purpose clustering (128 datasets)	include ED, DTW, others; deep models	Yes (84 clustering methods, 10 method classes)	tion of noise/misalignment/length	of elastic distances & deep models, but not factorized by mechanisms
[8] Paparrizos & Reddy 2023 (Odyssey)	Engine for general-purpose clustering evaluation	Supports various distances (ED, DTW, SBD, etc.) and methods	Yes (80 methods, 9 classes)	Infrastructure only; does not itself manipulate noise/misalignment/length	Experimental platform you could emulate or extend for your mechanistic study
[9] Forest et al. 2021	Clustering <i>validation</i> (model selection)	Agnostic to specific distance, but designed for time-series invariances	Applicable to many algorithms; not distance-specific	Explicit perturbations aligned with invariances (shift, warp, noise); stability as validation criterion	Directly relevant for experimental design to probe robustness to specific transformations, without prescribing the similarity measure
[10] Dove et al. 2023	Ecology; general time-series analysis tasks	42 distance measures; broad set including lock-step, elastic, others (not all paradigms elaborated in excerpt)	Clustering is one among several tasks	16 properties of distance measures , including noise sensitivity, phase invariance, etc.; evaluated per measure	Conceptually central for mapping distance “properties” to mechanisms (noise, misalignment, etc.) , though not clustering-centric or cross-paradigm in your sense

Key conclusion: None of these papers fully realize your desired combination of: (i) general purpose clustering, (ii) **all four paradigms** (lock-step, elastic, SBD-like, distributional/kernel/IDK), and (iii) **systematic factorized manipulation of noise, misalignment, and length**. Several, however, give strong partial coverage that can be combined into a mechanistic framework.

Comparative Coverage of Similarity Paradigms and Their Findings

Paradigm Coverage by Paper

The following table isolates which similarity families each paper actually compares, and whether this comparison is central.

Ref	Lock-step (ED or variants)	Elastic (DTW, MSM, etc.)	Sliding / SBD-like	Distributional / Kernel / Divergence	Is Cross-Paradigm Comparison a Main Aim?
[1]	Yes (ED, CID, etc.)	Yes (DTW, MSM, TWED, others)	Yes (k-sliding/SBD-like)	No explicit kernel/IDK; no divergence measures mentioned	Yes, among many distances, though framed around smart-meter clustering
[2]	Yes (ED)	Yes (DTW with fixed SC-band)	Yes (SBD / shape-based distance)	No	Yes, but only <i>three</i> paradigms (ED, DTW, SBD)
[3]	Yes (multiple)	Yes	Yes	Yes (4 kernel functions and 4 embedding measures)	Yes, but across distances and tasks; not focused on clustering alone
[4]	Yes (52)	Yes (7)	Yes (4)	Yes (4 kernels + 4 embedding)	Yes, but for <i>classification</i> rather than clustering
[5]	Yes (ED baseline)	Yes (9 elastic distances, focus on MSM, TWE, etc.)	No	No	No; comparison is intra elastic vs ED
[6]	Yes (ED) on distributions of differences	Yes (DTW) on distributions of differences	No	Yes (KL, Wasserstein on distributions)	Partly: they contrast distance types on the same <i>distributional representation</i>
[7]	Yes (ED)	Yes (DTW)	Likely yes (shape-based methods), but not detailed in excerpt	Some methods may implicitly be kernel-based; not the focus	No; focus is clustering methods, not distance paradigms per se
[8]	Yes (ED)	Yes (DTW)	Yes (SBD)	Some methods may use kernels; engine is extensible	No; Odyssey is infrastructure
[9]	Distance-agnostic	Distance-agnostic	Distance-agnostic	Distance-agnostic	No; focuses on validation criterion, independent of chosen distance
[10]	Yes	Yes		Possibly some kernel-like or dis-	

			Likely includes some shape-based measures	tributional measures among the 42	No; the goal is property-based selection for ecological tasks
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Implications:

- **Closest match to your four-paradigm scope:**— [3] and [4] are unique in explicitly including **kernel** and **embedding** measures *alongside* lock-step, sliding, and elastic distances on a large scale. However, their primary task is **classification**, not clustering.— [6] is the only paper that *explicitly* includes **distributional/divergence-based distances** (KL, Wasserstein) within a clustering setup, but in a very specific “distribution-of-differences” design, and without SBD-type measures.
- **Core cross-paradigm clustering benchmarks:**— [2] (ED vs DTW vs SBD) and [1] (ED + multiple elastic + SBD-like) jointly provide good coverage of **lock-step vs elastic vs sliding** in clustering context, though without kernel/IDK methods and without careful decomposition of mechanisms.
- **Elastic-only but high diagnostic value:**— [5] is focused on elastic distances for clustering, but its findings on **MSM vs DTW vs Euclidean** under k-means and k-medoids [5] are highly informative about **what goes wrong when the distance geometry mis-matches the clustering objective**.

Comparative Findings on Paradigm Strengths and Weaknesses

The table summarizes the **headline findings** about relative strengths of paradigms where the papers make clear, statistically supported claims.

Aspect	ED (lock-step)	Elastic (DTW, MSM, etc.)	Sliding/SBD	Distributional / Kernel
General accuracy in clustering benchmarks	In [2], ED is often competitive or superior to DTW for clustering on UCR datasets under their setup [2]. In [5], ED outperforms DTW in k-means clustering, largely due to centroid issues [5].	In [5], MSM consistently outperforms other elastic distances and DTW under k-medoids; DTW under k-means is often worse than ED because arithmetic means are poor prototypes for DTW geometry [5]. [1] finds DTW & k-sliding outperform “traditional” measures (ED, etc.) on smart-meter data [1].	In [2], SBD/shape-based similarity is competitive with ED and DTW across UCR clustering benchmarks [2]. In [1], k-sliding/SBD-like distances are among the top performers, especially when local shifts are prominent [1].	In [4], kernel distances do <i>not</i> universally dominate; they are competitive but not clearly superior across classification tasks [4]. In [3], kernel and embedding measures provide benefits in some multivariate scenarios but are not uniformly best [3]. In [6], KL and Wasserstein on difference distributions underperform ED and KL in both efficiency and clustering quality [6].
Sensitivity to misalignment (phase / local warping)	Several studies implicitly confirm that ED suffers when phase variation is present: in [1], methods allowing local temporal shifts (DTW, k-sliding) consistently outperform traditional lock-step measures for smart-meter patterns with timing variation [1]; [10] discusses phase sensitivity as an important property and notes ED is not shift-invariant [10].	DTW and related warping/edit distances address misalignment by construction. [1] reports that distances accommodating temporal shifts (DTW, MSM, TWED) are more robust to local timing variability in power consumption patterns [1]. [5] notes that <i>over-warping</i> and lack of regularization can hurt DTW in clustering, whereas MSM’s edit-based formulation is more stable [5].	SBD and k-sliding distances explicitly allow global shifts via normalized cross-correlation. [2] emphasizes that SBD is shape- and phase-invariant at the global level and performs well when <i>global shifts</i> dominate [2]. [1] finds SBD-like methods robust on smart-meter data where daily peak times may shift [1].	Kernel and distributional approaches in [3,4,6] are generally invariant to perturbations of time points (e.g., bag-of-features), or focus on spectral / autocorrelation properties, providing robustness to misalignment but potentially losing temporal order information. [6] emphasizes that using distributions of differences makes the distance focus on increment distributions, which are order-agnostic; this helps when only distributional differences matter, but may hurt when temporal patterning is key [6].
Interaction with clustering objective	ED is a true metric in Euclidean space, so arithmetic means are valid centroids. [5] shows that this compatibility explains ED’s strong performance with k-means relative to DTW [5].	Many elastic distances (especially DTW) are non-metric and live in non-Euclidean spaces. [5] empirically shows k-means with DTW performs poorly; switching to k-medoids or using DBA centroids improves DTW but doesn’t beat MSM [5]. MSM and TWE, being more metric-like under certain cost regimes, integrate better with prototype/medoid clustering [5].	SBD-based k-shape and related methods use shape-based centroids defined via cross-correlation, avoiding Euclidean centroid assumptions. [2] discusses k-Shape’s behavior on SBD and observes that shape-based centroids often outperform standard k-means centroids for misaligned data [2].	Kernel measures enable kernel k-means , where centroids are means in feature space. [4] shows that some kernels are PSD and can be used in standard kernel methods; however, empirical results suggest that carefully tuned non-kernel distances can match or beat kernel-based ones [4]. [3] reinforces that kernel/embedding measures do not universally dominate for multivariate tasks [3].
	[4] shows that normalization choice significantly impacts performance , and z-normalization is not always best; this extends common wisdom	Elastic distances are strongly impacted by normalization and parameter choices (e.g., window size, warping penalties). [1] observes sensitivity of optimal	SBD typically uses z-normalization and correlation-based scaling. [2] imposes uniform pre-processing (z-normalized, no extra filters), helping avoid	Kernel methods rely heavily on hyperparameters (bandwidths, decay factors, etc.). [4] reports that some kernel measures are quite sensitive to param-

Effect of normalization & preprocessing	about ED [4]. [3] confirms and extends these findings to multivariate distances and multiple normalization schemes [3].	DTW window to dataset properties [1]; [5] stresses that parameter tuning is critical and that unregularized DTW often over-warps [5].	confounds [2]. However, [4] notes that alternative normalizations can lead to different relative performances among sliding distances [4].	ter choices and that failure to tune them can obscure their potential [4]. [3] mirrors this for multivariate kernels. In [6], divergence distances on difference distributions depend on how distributions are estimated, which affects robustness.
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Takeaway for your study:

- **Elastic vs lock-step:** [5] and [1,2] jointly show that **elasticity helps when misalignment is present** but can **hurt clustering when combined with inappropriate centroids or poorly tuned warping costs**.
- **SBD vs DTW:** [2] and [1] suggest SBD-type methods can match or beat DTW on clustering tasks where **global phase variation** dominates; they are cheaper and easier to integrate with centroid-based methods.
- **Distributional/kernels:** [3,4,6,10] reveal **no consistent superiority** of kernel/distributional measures but highlight their distinctive invariance properties (e.g., order-invariance, spectral sensitivity), which you can exploit in mechanistic experiments.

Treatment of Noise, Temporal Misalignment, and Sequence Length

Which Papers Explicitly Manipulate or Theorize About These Factors?

Ref	Noise Level Explicitly Varied?	Temporal Misalignment Explicitly Varied/Modeled?	Sequence Length Explicitly Varied/Analyzed?	Notes
[1]	Yes – synthetic datasets vary noise and outlier levels, cluster balance, and dataset composition [1].	Partly – generative design introduces local temporal shifts and warping in peak consumption, but misalignment is not parameterized as a clean factor sweep [1].	No explicit length manipulation; series lengths are set by domain (e.g., daily/weekly load profiles).	One of the few clustering studies to systematically vary noise and dataset properties , though designed around smart-meter domain.
[2]	No – real UCR datasets with inherent but uncontrolled noise.	No – misalignment is whatever the dataset naturally has; no synthetic warping experiments.	All datasets are equal-length; no factorized analysis of length [2].	Focus on clean benchmarking across algorithms and distances, not on perturbation mechanisms.
[3]	No explicit manipulation; examines normalization and channel dependence but not noise per se.	No explicit warping/shift experiments; misalignment emerges naturally from datasets.	Not the focus; multivariate lengths vary but are not a controlled variable [3].	Strength is in breadth of distance measures and normalization, not perturbations.
[4]	No controlled noise variation; uses 128 datasets with their inherent noise.	Misalignment is addressed implicitly via inclusion of sliding and elastic measures, but no experiments where misalignment is gradually increased [4].	Sequence lengths vary across datasets, but not systematically analyzed.	Focus on debunking misconceptions via large-scale classification experiments and normalization analysis.
[5]	No – all UCR datasets used as-is; no synthetic noise injection.	No controlled misalignment; UCR datasets do include misalignment, but it's not parametrically varied [5].	Equal-length series; effect of length is not discussed beyond computational cost.	Core insight: geometry/clustering-objective mismatch dominates performance more than noise or misalignment in their setup.
[6]	No controlled noise; robustness inferred indirectly via differencing and distributional distances [6].	Not explicitly; differencing may reduce some low-frequency misalignment effects, but not systematically studied.	No explicit sequence-length analysis.	Preliminary, small-scale study with a distributional viewpoint .
[7]	No explicit perturbation design; uses 128 datasets [7].	Misalignment is handled implicitly by underlying distances and methods, but not factorized as an experimental variable.	Length varies across datasets but is not isolated as a factor.	A breadth-oriented benchmark rather than mechanism-oriented.
[8]	No inherent perturbation study; Odyssey is an engine that could be used to run such experiments [8].	N/A	N/A	Methodological infrastructure.
[9]	Yes, by design – stability criterion uses perturbations aligned with invari-	Yes – they explicitly consider shift and warp invariances and define per-	Potentially yes – subsampling/reshaping perturbations can impact effective	Provides a model-agnostic framework for probing robustness to spec-

	ances, which can include additive noise [9].	turbation processes accordingly [9].	length, but not discussed as a primary factor.	ified transformations and selecting k.
[10]	Yes, conceptually – among the 16 properties, they test noise sensitivity and related invariance properties for each distance [10].	Yes, conceptually – properties include sensitivity to phase shifts, warping, and other temporal transformations [10].	Length effects are not central, but some properties implicitly depend on series length (e.g., scaling with dimensionality).	Key contribution is property-based characterization of distances relevant to noise/misalignment, rather than direct clustering experiments.

Synthesis:

- **Noise:**– [1] and [9] are the only works clearly designing **controlled perturbation experiments** for clustering/validation with explicit noise variation.– [10] provides a systematic *property-based* evaluation of noise sensitivity across 42 distances, but outside a clustering evaluation.
- **Temporal misalignment:**– [9] is unique in formalizing **warp and shift invariances** as perturbations for stability-based model selection.– [1] introduces misalignment in synthetic smart-meter data but does not isolate it as a separate sweepable parameter.– [10] empirically marks which distances are invariant or sensitive to particular temporal transformations, giving a property map you can re-use.
- **Sequence length:**– **None** of these works conducts a systematic, controlled study of how **sequence length** affects relative performance of similarity paradigms in clustering, beyond anecdotal or computational comments. This is a clear **gap and opportunity** for novelty in your study.

Comparative Experimental Design Choices Relevant to Mechanistic Analysis

Control of Confounding Factors (Preprocessing, Parameters, Algorithms)

Ref	Preprocessing Control	Parameter Tuning Strategy	Clustering Algorithms & Objectives
[1]	Mixture of real and synthetic smart-meter series; domain-informed normalization (e.g., peak emphasis), but not as standardized as UCR-style benchmarks [1].	Some parameter tuning; they note sensitivity of DTW window and use random draws plus focused tuning for robustness [1].	11 clustering algorithms including hierarchical (Ward's linkage), partitioning, and others, with distance plugged in [1]. Focus on robustness across methods.
[2]	Strong control: all 112 UCR datasets, equal length, no additional preprocessing beyond standardized normalization; same preprocessing across distances and algorithms [2].	DTW uses fixed 5% Sakoe–Chiba window following prior work; parameters otherwise kept consistent to avoid overfitting [2].	8 clustering methods (k-means, k-medoids, hierarchical, density-based, etc.), consistent across distances [2].
[3]	Multivariate datasets with systematic exploration of 13 normalizations and 2 channel-dependency models [3].	Extensive parameter sweeps for distance-specific hyperparameters in both supervised and unsupervised settings [3].	Includes clustering as one task, but not the sole focus. Objective is to disentangle normalization and multivariate coupling effects .
[4]	8 normalization methods tested across 71 distances on 128 datasets [4].	Parameters tuned via cross-validation or statistically sound protocols; extensive statistical tests (e.g., Nemenyi) used [4].	Classification-focused; clustering algorithms are not central. Experimental design emphasizes fair, statistically validated comparison of distances .
[5]	All UCR univariate datasets, equal-length and z-normalized [5].	Default or recommended parameters for elastic distances; some critical discussion of parameter choices but not exhaustive sweeps [5].	k-means and k-medoids with multiple initializations; also consider DBA for DTW centroids [5].
[6]	Time series transformed into distributions of first-order differences ; no mention of normalization beyond differencing [6].	Parameters mainly for KL / Wasserstein estimation; not systematically tuned across a wide grid.	Graph partitioning clustering on the resulting distance graph [6].
[7]	Uses the same 128 datasets as [4] with standardized preprocessing (as per Odyssey/aeon setup) [7].	Parameters tuned per method class; particular attention to deep models and their hyperparameters [7].	84 clustering methods spanning 10 classes (including classical, kernel, deep), evaluated under consistent metrics with statistical tests [7].
[8]	Encodes preprocessing and normalization as configurable modules; uses the same datasets/preprocessing as [4,7] by default [8].	Supports parameter sweeps and reproducible configurations; tuning strategy depends on user [8].	Provides a uniform environment for k-means, k-medoids, hierarchical, density-based, deep clustering, etc. [8].
[9]	Preprocessing left to user; they assume the data reflect some invariances that can be encoded as perturbations [9].	Focus is on perturbation design rather than distance parameters.	Clustering algorithm-agnostic; stability criterion applies broadly to many algorithms and distances [9].

[10]	Preprocessing not the central variable; series are used to evaluate distance properties under controlled manipulations [10].	No clustering-specific tuning; each distance is assessed on properties via controlled tests [10].	Many tasks considered (classification, clustering, etc.), but experiments mainly illustrate distance selection in ecological applications [10].
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Design lessons for your study:

- **From [2,3,4,7,8]:** adopt **strict control over preprocessing and normalization**, and perform statistically grounded comparisons; ensure that distance comparisons are not confounded by ad hoc preprocessing.
- **From [5]:** explicitly align the **clustering objective** with the geometry of each distance (e.g., use medoids for non-Euclidean distances, barycenters for DTW), and analyze performance differences that emerge when this alignment is broken.
- **From [1,9,10]:** design **synthetic perturbation experiments** where *noise level*, *misalignment magnitude*, and *length* are directly controlled and varied across regimes; tie perturbations to clear invariance assumptions.

Comparative Conceptual Contributions Relevant to a Mechanistic Framework

Conceptual / Theoretical Foundations

- **Unified view on distance properties and misconceptions**– [4] debunks four misconceptions about time-series distances, including simplistic beliefs about DTW always outperforming ED, and z-normalization always being best. It provides the most thorough **taxonomy and statistical evaluation of distance measures** across paradigms [4].– [10] systematizes **16 properties** of distances (including sensitivity to noise, phase shifts, warping, outliers, etc.) and tests 42 measures against them [10]. This turns intuitive statements about robustness and invariance into **empirically checked properties**, very useful for mechanism-based modeling.
- **Multivariate and normalization aspects**– [3] shows that conceptual insights from univariate distances largely carry over to **multivariate series**, but that normalization choices and channel-dependency modeling significantly affect performance [3]. This underscores that your study must carefully define **how multivariate similarity is constructed** (independent vs joint channel structure).
- **Clustering-geometry mismatch**– [5] provides a clear empirical argument that **distance geometry and clustering objective have to be aligned**. DTW performs poorly with k-means but improves with k-medoids and DBA; MSM's metric-like structure makes it more amenable to clustering [5]. This is essential for interpreting your cross-paradigm results mechanistically, rather than as artifacts of inappropriate prototype definitions.
- **Robustness and stability as mechanisms**– [9] frames clustering quality in terms of **stability under perturbations consistent with known invariances** (shift, warp, noise), providing a generic lens to evaluate how well a chosen distance measure and clustering method reflect the assumed invariances [9].
- **Distributional vs alignment-based perspectives**– [6] takes a distributional approach by comparing **distributions of first-order differences** with divergences (KL, Wasserstein) and ED/DTW [6]. Although small-scale, it illustrates the **mechanistic shift** from pointwise/warping alignment to viewing time series as samples from a distribution over increments, which is closely related to **IDK-type methods** and kernel mean embeddings.

Where Your Study Can Be Novel

Combining the above:

- No paper **simultaneously**:
 1. Compares **ED, elastic (DTW, MSM), SBD-type, and distributional/kernel/IDK methods** in a **clustering** context;
 1. Uses **synthetic or semi-synthetic data** where **noise level, temporal misalignment (global shift and local warping), and sequence length** are *independently controlled* and **systematically swept**;
 1. Aligns the **clustering objective** (means vs medoids vs barycenters vs kernel means) with each similarity measure's geometry; and
 1. Provides **mechanistic explanations** grounded both in empirical perturbation responses and in distance properties from [4,10].
- The closest building blocks are:
 - **Cross-paradigm distance benchmarking**: [2,3,4,7,8]
 - **Elastic vs ED geometry in clustering**: [5]
 - **Noise & misalignment perturbations / invariance-based validation**: [1,9,10]
 - **Distributional / divergence-based distances**: [6]
 - **Distance property taxonomies and normalization effects**: [3,4,10]

Your work can synthesize these strands by:

- Designing **factorial experiments** in which:
 - Noise level, misalignment (global shift vs local warp), and sequence length are factors;
 - Each similarity paradigm is **instantiated with best-practice parameterization** (informed by [3,4,5]);

- The clustering objective is appropriately matched (Euclidean k-means for ED, k-medoids/MSM for elastic, k-Shape/SBD-centroids for SBD, kernel k-means for distributional kernels).
- Using the **distance property maps** from [4,10] and the **stability framework** from [9] to interpret your empirical curves as **manifestations of specific invariance and robustness mechanisms**, instead of treating results as black-box accuracies.

This comparative analysis should give you a precise sense of which aspects of your intended mechanism-oriented, cross-paradigm study are supported by existing literature and which dimensions—particularly **sequence length effects** and **IDK-like distributional kernels for clustering under controlled perturbations**—are open for genuinely novel contributions.

Timeline

Chronological Evolution of Cross Paradigm Time Series Similarity and Clustering

Pre 2020 context (implied background)

Although none of the provided papers are pre 2020, several of them explicitly position themselves against an older, now canonical body of work:

- Paparrizos et al. [4] repeatedly refer to a “most detailed experimental study” on time series distances that is “over a decade old” and **univariate, classification centric, ED vs. DTW centric, and limited in normalization and measure variety**
- Post 2020 works consistently build on the **UCR archive** tradition (fixed length, mostly univariate datasets; supervised benchmarks), and on **DTW vs. ED** as the default comparison paradigm.

This pre 2020 landscape can be summarized as:

- Heavy emphasis on **classification**, not clustering.
- Focus on **lock step vs. elastic**(ED vs. DTW and some variants).
- Limited treatment of **sliding/shape based** or **kernel/distributional** distances.
- **Mechanisms** such as noise, misalignment, and sequence length effects are mostly implicit, not systematically isolated.

The post 2020 developments can be read as a response to this gap.

2020: Systematic Distance Studies and the First Clustering Benchmark

2020 – “Debunking Four Misconceptions” as a distance measure reset [4]

Paparrizos et al. [4] provide a **large, classification oriented bake off** over 128 datasets, with:

- **71 distance measures:**
 - 52 **lock step**
 - 4 **sliding** (SBD type)
 - 7 **elastic** (DTW and relatives)
 - 4 **kernel functions**
 - 4 **embedding measures**
- **8 normalization methods.**

Key contributions to the **timeline of ideas**:

- **Broadening the distance landscape:**
 - Goes beyond ED vs. DTW to a *structured taxonomy* of lock step, sliding, elastic, kernel, and embedding distances.
 - Introduces **multiple sliding and kernel measures** into the same experimental frame, which is an early cross paradigm comparison (albeit in supervised classification).
- **Strong emphasis on experimental rigor:**
 - Statistical tests, normalization analysis, and controlled settings.
 - This **benchmarking culture** (large datasets, many distances, careful statistical analysis) becomes the template for later clustering work [2,3,7,8].
- **Conceptual impact:**
 - Explicitly **debunks misconceptions** (e.g., “DTW is always superior to ED”) and shows that **normalization choice and distance-task fit** matter as much as the distance family itself.

- Provides a **taxonomy and empirical map** that later clustering work builds on, even when focusing on fewer measures (ED/DTW/SBD).

While [4] is classification centric, it is the **first modern, large scale cross paradigm distance evaluation** across lock step, sliding, elastic, and some kernel measures, and it heavily influences subsequent clustering studies [1,2,3,7,8,10].

2020 – First large clustering benchmark over UCR: Javed et al. [2]

Javed et al. [2] perform one of the first **systematic time series clustering benchmark** over the UCR archive (112 datasets), evaluating:

- **8 clustering algorithms** (partitional, hierarchical, density based).
- **3 distance paradigms**:
 - **ED** (lock step)
 - **DTW** (elastic; with a fixed 5% Sakoe–Chiba window)
 - **SBD** (shape based, sliding/shift invariant).

Design choices and contributions:

- **Focus on clustering**, not classification; but keeps:
 - Equal length series.
 - No extra preprocessing.
 - Known number of clusters (from labels).
- Explicit comparison of **ED vs. DTW vs. SBD** gives one of the first **cross paradigm cluster level** evaluations.

Limitations relative to a mechanism oriented study:

- No explicit manipulation of **noise level**, **misalignment amplitude**, or **sequence length** as controlled variables.
- No **distributional/kernel** methods beyond SBD.

Still, [2] is a clear milestone: it transforms the **distance bake off mindset** [4] into the **clustering domain**, setting a baseline for later, broader clustering benchmarks [5,7,8].

2021: Stability and Model Selection under Perturbations

2021 – Invariance guided stability for clustering validation [9]

Forest et al. [9] introduce a **stability based model selection criterion** tailored to time series:

- Key idea: choose clusterings that are **stable under perturbations** consistent with known **invariances** (shift, warping, noise).
- They combine **between cluster** and **within cluster** stability and show how **invariance assumptions** guide the perturbation process.

Impact on the conceptual timeline:

- Shifts attention from **distance choice only** to the **interaction between invariance assumptions and stability**.
- Offers a **mechanistically oriented validation approach**:
 - Perturbations explicitly simulate **noise** or **misalignment** according to hypothesized invariances.
 - Stability becomes a proxy for robustness w.r.t. these mechanisms.

While [9] does not itself compare ED/DTW/SBD/kernels, it provides **experimental design and validation tools** that are directly relevant to mechanism oriented studies of how **noise and misalignment** affect clustering, and how different distances encode invariance.

2022: Elastic distance bake off and first distributional style clustering

2022 – Elastic distances for clustering: Holder & Bagnall [5]

Holder et al. [5] perform a **large empirical study focused exclusively on elastic distances** for partitional clustering (k means and k medoids) over the univariate, equal length UCR archive:

- Distances: **DTW, DDTW, WDTW, ERP, EDR, LCSS, TWE, MSM, WDDTW** plus ED.
- Clustering algorithms: **k means (with Euclidean mean)** and **k medoids** plus DBA based variants for DTW centroids.
- All series are **z normalized**

Key findings:

- With k means, **DTW often underperforms ED**; with k medoids, all elastic distances improve.
- **MSM** emerges as consistently strong; **TWE** close second.
- **Hybrid editing + warping** distances (MSM, TWE, ERP, WDTW) tend to outperform pure DTW, suggesting a **different inductive bias** and more metric like behavior.
- Highlights issues with **centroid definition and optimization** when using elastic distances.

Conceptually important points:

- This is a **deep dive into the elastic family**, spotlighting **MSM/TWE vs. DTW** in unsupervised clustering.
- Contributes detailed **mechanistic insights** about:
 - Why over warping DTW can hurt clustering.
 - Why metric like properties (MSM) help k medoids.
- However, it is **single paradigm**(elastic vs ED baseline) and **does not** bring in **sliding/SBD** or **distributional/kernel** methods, nor does it isolate **noise/misalignment/length** as explicit experimental factors.

For the field, [5] reinforces the idea that “**DTW is not the last word on elastic distances**” and shifts attention toward **MSM and related edit warp hybrids** as strong clustering distances.

2022 – Distribution over differences with multiple distances: Thanakulkairid et al. [6]

Thanakulkairid et al. [6] propose clustering based on the **distribution of first order differences** of the time series, then compare:

- **ED, Kullback–Leibler divergence, Wasserstein distance**, and **DTW** on these *difference distributions*
- They model each time series as a **graph vertex** with edge weights given by these distances; then perform **graph partitioning**.

Contributions to the timeline:

- Introduces a **distributional perspective** on time series differences in a clustering context.
- Compares a **classic lock step measure (ED)** an **elastic measure (DTW)**, and **two distributional/probability based distances** (KL, Wasserstein).

In relation to the requested **IDK/distributional kernel paradigm**:

- It is not a kernel method per se, but it clearly **moves towards distributional comparisons** of time series by treating their **difference sequences as empirical distributions**.

This paper is a **small but important step** toward integrating **distributional similarity** into general purpose time series clustering, beyond the usual ED/DTW/SBD triad.

2023: Domain specific guidance and benchmarking infrastructure

2023 – Ecology focused guide to distance measures: Dove et al. [10]

Dove et al. [10] curate **42 distance measures** and classify them along **16 properties** that matter to ecological time series tasks; they then:

- Provide an **objective procedure** for selecting distance measures based on desired properties (e.g., sensitivity to phase shifts, noise, scaling).
- Demonstrate the selection method on ecological clustering and related tasks.

Relevance and impact:

- Translates the **computer science distance zoo**(including many measures from [4]) into a **domain oriented, property driven framework**.
- Emphasizes **properties like shift invariance, warping sensitivity, noise sensitivity**, etc., which are essentially the **same mechanisms** (misalignment, noise) the mechanism oriented study is interested in.

While not a cross paradigm clustering benchmark, [10] underscores a **property based, mechanistic view of distances** and shows that **task aligned choice of invariances** is critical.

2023 – Odyssey: infrastructure for clustering benchmarking [8]

Paparrizos & Reddy [8] introduce **Odyssey**, a web engine to benchmark **80 time series clustering methods from 9 classes** over 128 datasets, with statistical analysis.

Key contributions to the research timeline:

- **Infrastructure and standardization**:
 - Odyssey provides a **modular, extensible platform** to evaluate clustering methods using multiple distance measures, algorithms, and datasets.
 - Enables **reproducible, large scale experiments** and systematic parameter tuning.
- **Broad but method centric view**

- Focuses on **clustering method classes** (including deep learning) rather than on **distance paradigms per se**.
- Nonetheless, it **challenges claims about elastic measures in clustering** and highlights **parameter tuning effects**.
- Conceptual significance:
 - Signals a shift toward **continuous, community level benchmarking** rather than isolated one off studies.
 - Provides a **testbed** where cross paradigm distance analyses—including those involving **distributional kernels**—could be integrated.

Odyssey [8] is less about new theory and more about **evaluation infrastructure**, but it is a major **enabler** for future cross paradigm, mechanism oriented studies.

2024–2025: Robustness to dataset properties and multivariate distances

2024 – Smart meter clustering with distance and dataset property analysis: Yerbury et al. [1]

Yerbury et al. [1] study smart meter time series clustering in three phases:

1. **31 distance measures** and **8 representation methods** assessed via leave one out classification.
1. Selection of better performing methods and clustering with **11 algorithms**.
1. **Robustness experiments** where **dataset properties** (cluster balance, noise, outliers) are systematically varied on **expert informed synthetic data** with peak consumption patterns.

Distance paradigms explicitly include:

- **ED baselines**.
- Several **elastic** methods: DTW (default and constrained), MSM, TWED, etc.
- **Alignment/sliding** methods, including **k sliding/SBD**
- Other measures like MPD, CID.

Key conceptual advances:

- One of the first studies to **explicitly and systematically vary dataset properties** (noise, cluster balance, outliers) while holding other factors controlled.
- Reports that **methods allowing local temporal shifts while preserving amplitude—e.g., DTW and k sliding/SBD—are consistently strong**, and that **constraining the DTW window** improves both performance and efficiency.
- Shows that combinations of such distances with **hierarchical clustering (Ward's linkage)** are robust, even with randomly drawn parameters.

In relation to the **mechanism oriented goal**

- [1] is the **closest among the listed works** to a mechanism oriented design:
 - Uses **synthetic data with controlled characteristics**.
 - Explicitly investigates how **noise and dataset composition** affect robustness.
 - Compares **multiple paradigms** (lock step, elastic, sliding); however, **distributional/kernel methods are absent**, and **sequence length** is not systematically varied.

This paper marks a shift from **pure benchmarking** to more **experimental control** over generative factors—a significant milestone for the kind of study you aim to conduct.

2025 – Multivariate distance measures: d'Hondt et al. [3]

d'Hondt et al. [3] present a **structured study of multivariate time series distances** evaluating:

- **30 standalone measures** across **8 categories**, with **2 channel dependency models** and **13 normalizations**.
- **3 downstream tasks** (not fully detailed in the excerpt, but including unsupervised tasks).
- They explicitly note that previous work:
 - Focused mainly on **lock step and elastic** distances.
 - Mostly ignored **sliding** and **kernel** categories, especially in the multivariate setting.

Contributions:

- Extends the **taxonomy-based, multi category view from [4]** into the **multivariate** domain.
- Explicitly includes **sliding and kernel measures** among the 8 categories, addressing exactly the prior omission they highlight.
- Provides **rigorous parameter search** and **statistical analysis** to ensure fair comparison.

Relevance for cross paradigm, mechanism oriented work:

- [3] gives a **rich catalogue of multivariate distances** spanning:

- **Lock step**
- **Elastic**
- **Sliding**
- **Kernel** plus other categories.
- However, like [4], its focus is on **overall performance and normalization** rather than on controlled variation of **noise, misalignment, and length**.
- It does, nonetheless, **confirm that insights from univariate distances largely transfer to multivariate data**.

Overall, [3] marks a **broadening from univariate to multivariate**, and reinforces **kernel and sliding measures** as first class citizens in large scale evaluations.

2025 – Comprehensive clustering benchmark across method classes: Paparrizos & Teja [7]

Paparrizos & Teja [7] extend Odyssey style benchmarking:

- Evaluate **84 time series clustering methods** across **10 method classes**, including **data mining, machine learning, and deep learning**.
- Use **128 datasets** and rigorous statistical methods.
- Goals include:
 - Identifying **top methods per class**.
 - Highlighting **overlooked high performing classes**
 - **Challenging assumptions about elastic measures**.
 - **Refuting claimed superiority of deep learning methods, including foundation models**.
 - Exposing weaknesses in existing evaluation setups.

In the timeline:

- [7] is the **clustering counterpart** of [4]: a broad, multi class empirical map, but focused on clustering methods rather than just distances.
- Builds explicitly on [2,3,4,8], suggesting a sustained line of work from distance benchmarking through Odyssey to comprehensive clustering evaluation.

For the mechanism oriented agenda:

- [7] exposes the **overall landscape** of clustering methods under a common benchmark, showing that:
 - Elastic measures are **not uniformly superior**.
 - Deep methods do not automatically dominate.
- However, like [2], it does not explicitly manipulate **noise, misalignment, or length** as independent variables.

Collaborator Clusters and Sustained Research Threads

Paparrizos and collaborators: distance benchmarking 'Odyssey' comprehensive clustering

Key works: [4,2,8,3,7]

- **[4] (2020)**: Broad distance measure benchmark (classification), establishing a **taxonomy and experimental template**.
- **[2] (2020)**: First major **clustering** benchmark over UCR with ED, DTW, and SBD.
- **[8] (2023)**: **Odyssey** as benchmarking infrastructure for clustering.
- **[3] (2025)**: **Structured multivariate distance study**, including sliding and kernel measures.
- **[7] (2025)**: **Comprehensive clustering benchmark** across 10 method classes, including deep learning.

Sustained contributions:

- Promote a **benchmark driven, statistically rigorous culture**
- Systematically broaden from:
 - Univariate to **multivariate**.
 - Classification to **clustering**.
 - Simple measure sets (ED/DTW) to **rich taxonomies** including sliding and kernel methods.
- Challenge "**DTW superstition**", and later, **deep learning myths** in clustering.

For your purposes, this group's work forms the **empirical backbone** of what is known about cross paradigm distances and clustering, but **without** mechanistic noise/misalignment/length experiments or deep **distributional kernel** (IDK style) treatments.

Bagnall and colleagues: elastic distances and aeon toolkit

Key work: [5]

- Focused on **elastic distance functions** and their performance in partitional clustering.
- Introduces and evaluates **MSM and related edit warp hybrids** as strong competitors to DTW.
- Opens discussions about **prototype definitions (barycenters)** and the practicalities of using non Euclidean distances with k means.

Contribution to the timeline:

- Provides a **counter narrative to DTW dominance** and strengthens the case for **MSM/TWE** in clustering.
- Through the **aeon** toolkit, contributes **software infrastructure** that many later experimental studies can leverage.

Application and property oriented threads: Yerbury [1], Dove [10], Forest [9], Thanakulkairid [6]

These works, though not forming a single author cluster, each contribute distinctive thematic threads:

- **Yerbury et al. [1]:**
 - Domain: **energy/smart meters**.
 - Contribution: tie **cross paradigm distances** (ED, DTW, MSM, SBD) to **dataset properties** (noise, outliers, cluster balance) in a semi synthetic setting.
 - Represents **the most mechanistic clustering study** among the listed works.
- **Dove et al. [10]:**
 - Domain: **ecology**.
 - Contribution: **property based selection** of distances (sensitivity to shift, warping, noise), bridging theoretical properties to ecological needs.
- **Forest et al. [9]:**
 - Methodological focus: **stability based model selection using invariance guided perturbations** directly relevant to evaluating robustness w.r.t. noise/warp/shift.
- **Thanakulkairid et al. [6]:**
 - Methodological focus: **distribution of first order differences** and comparison of **ED, DTW, KL, Wasserstein**, pushing toward **distributional similarity** in clustering.

Together, these threads suggest a **shift from raw performance to property and invariance aware design and evaluation** which aligns closely with a mechanism oriented perspective.

Emerging Patterns and Gaps Relevant to Mechanism Oriented Cross Paradigm Studies

1. Trend: From single paradigm ED/DTW to multi paradigm taxonomies

- Early modern work emphasizes **ED vs. DTW**; by 2020, [4] and [2] bring in:
 - **Sliding/SBD** distances.
 - A few **kernel functions**.
- Multivariate extension [3] explicitly adds **sliding** and **kernel** categories into the evaluation.
- Domain and application studies [1,10] pick up on this wider universe of distances.

Yet, **distributional kernels/IDK style methods** remain **underrepresented**:

- [4,3] include only **a small number of kernel functions**, mostly evaluated in classification or generic performance terms.
- Thanakulkairid et al. [6] is one of the few to explicitly use **distributional distances** (KL, Wasserstein) in clustering, but not kernels.

Implication: A mechanism oriented study that gives **distributional/IDK kernels equal footing** alongside **ED/DTW/MSM/SBD** is **largely novel** within the clustering literature represented here.

2. Trend: Benchmarking culture and statistical rigor

From [4] through [2,5,3,7], and [8], we see:

- Increasing:
 - **Number of datasets** (up to 128).
 - **Number of distances/methods**.
 - Use of **statistical tests** and **systematic parameter searches**.
- Development of **tooling and infrastructure**:

- **aeon** (via [5]) for distance implementations.
- **Odyssey** [8] for large scale clustering evaluations.

Implication: Your mechanism oriented study can **leverage these toolkits and principles**, but should **diverge** by designing **controlled generative experiments** (for noise, misalignment, length) rather than relying solely on benchmark archives.

3. Trend: Growing attention to invariance, normalization, and properties

- **Normalization and invariance:**
 - [4,3] examine multiple normalizations and note their **statistically significant impact**.
 - [10] codify distance measures in terms of **16 properties**, many related to **invariance and robustness**.
 - [9] explicitly uses **invariance assumptions** to guide perturbations in stability analysis.

This reflects a shift from “which distance wins on average?” to “which distance has the **right invariances for this problem?**”

Implication: A mechanism oriented comparison that manipulates **noise, temporal misalignment, and sequence length** is a **natural next step** in this property focused trajectory, but it has not yet been carried out comprehensively across **lock step, elastic, sliding, and distributional kernel** paradigms.

4. Trend: From univariate to multivariate and from distances to method classes

- **Univariate ' Multivariate**
 - [4,2,5] are primarily univariate (UCR).
 - [3] extends the analysis to **multivariate time series**, maintaining a multi category distance taxonomy.
- **Distances ' Full method classes (including deep learning)**
 - [5] focuses on **distances + k means/k medoids**
 - [7,8] broaden to **whole method classes**, including deep learning and foundation models, and show **no automatic superiority** for deep methods.

Implication: Your focus is **not** on deep architectures, but on **similarity paradigms**; the existing trajectory provides context that **distance choice remains crucial**, even in a deep learning era, and **multivariate extensions** are increasingly important.

5. Gap: Systematic, cross paradigm analysis of noise, misalignment, and length

Across the corpus:

- **Noise:**
 - Explicitly manipulated in [1] (synthetic smart meter data) and [9] (stability perturbations).
 - Implicit in benchmark diversity elsewhere, but rarely parameterized.
- **Temporal misalignment:**
 - Often discussed conceptually (e.g., DTW vs. ED vs. SBD), but not **varied as a controlled parameter** across paradigms.
- **Sequence length:**
 - Mentioned as a complexity issue; but **very few controlled studies** systematically vary length to probe distance behavior.

No paper among [1,2,3,4,5,6,7,8,9,10] conducts a **single, mechanism oriented experiment** where:

- **Multiple paradigms** (ED, DTW, MSM, SBD, distributional/IDK kernels) are evaluated under:
- **Controlled and orthogonal variations** of:
 - Noise level.
 - Temporal misalignment (phase shift, local warping).
 - Sequence length.

And then **interprets results mechanistically** in terms of the distances' inductive biases.

Yerbury et al. [1] are the closest, but:

- They do not cover **distributional/kernel** methods.
- They emphasize **dataset composition (noise, balance, outliers)** rather than **fine grained temporal misalignment and length**.

Implication: There is a clear **open niche** for the kind of **mechanism oriented, cross paradigm clustering study** you are proposing. It would be:

- **Aligned with current trends** (benchmarking, invariance, multivariate extensions).
- Clearly **distinct and novel** in its **explicit manipulation** of generative factors and inclusion of **distributional/IDK kernels** as equal players.

Looking Forward: What the Timeline Suggests About Future Directions

Based on the trajectory of these works, several likely future directions—and how your work fits—emerge:

1. From empirical bake offs to explanatory, mechanism driven studies

- Benchmarking has matured ([4,2,5,3,7,8]); **incremental bake off** have diminishing returns.
- The next step is **explanatory experiments** that link **distance properties** to **data generating mechanisms** (noise, misalignment, length), exactly as you propose.

1. Deeper integration of distributional and kernel paradigms

- Kernel measures appear in [4,3], and distributional distances (KL, Wasserstein) in [6], but they are not central in clustering.
- A study that **elevates distributional/IDK kernels** to **first class comparators** alongside ED/DTW/MSM/SBD would fill a conspicuous gap.

1. Invariance aware validation and model selection

- Methods like [9] show how **stability under invariance consistent perturbations** can guide model selection.
- Combining such criteria with **systematic noise/misalignment experiments** could lead to **practical tools** for choosing distances and clustering methods.

1. Application grounded synthetic frameworks

- Yerbury et al. [1]'s **expert informed synthetic data** for smart meters suggests a pattern:
 - Build synthetic datasets **anchored in real domains**, but with **controlled factors**.
- Your study could emulate this, but with **generic time series motifs** and **explicitly parameterized misalignment and noise models**, feeding results back to benchmarks via tools like **Odyssey** [8].

In summary, the field has progressed from **ED vs. DTW folklore** to **large scale, multi category, statistically rigorous benchmarks** and an increased awareness of **invariance and normalization**. However, it still lacks a **mechanism oriented, cross paradigm comparison including distributional/IDK kernels**, with **noise, misalignment, and length** treated as **first class experimental factors**. Your proposed work would be a **natural and novel next step** in this timeline.

Foundational Work

Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

Ref.	Reference Rate	Title	Cited By These Relevant Papers
[3]	1.00	A Structured Study of Multivariate Time-Series Distance Measures	[7]
[12]	1.00	A Survey on Time-Series Distance Measures	[3, 7]
[134]	0.78	Dynamic programming algorithm optimization for spoken word recognition	[1, 2, 3, 4, 5]
[22]	0.78	Experimental comparison of representation methods and distance measures for time series data	[1, 2, 3, 4, 7, 10]
[4]	0.76	Debunking Four Long-Standing Misconceptions of Time-Series Distance Measures	[1, 3, 7, 8, 10, 12]
[141]	0.67	Time-series clustering - A decade review	[1, 2, 5, 7, 8]
[46]	0.66	The Move-Split-Merge Metric for Time Series	[1, 3, 4, 5, 7]
[52]	0.66	Time Warp Edit Distance with Stiffness Adjustment for Time Series Matching	[1, 3, 4, 5, 7]
[51]	0.62	On The Marriage of Lp-norms and Edit Distance	[1, 3, 4, 5, 10]
[142]	0.61	A New Time Series Similarity Measure and Its Smart Grid Applications	[1]
[143]	0.61	On the Use of Relative Validity Indices for Comparing Clustering Approaches	[1]

[2]	0.60	A Benchmark Study on Time Series Clustering	[1, 5, 7, 8, 14]
[5]	0.57	A review and evaluation of elastic distance functions for time series clustering	[1, 8, 14]
[144]	0.55	On the Need for Time Series Data Mining Benchmarks: A Survey and Empirical Demonstration	[1, 2, 7, 14]
[93]	0.53	Robust and fast similarity search for moving object trajectories	[1, 4, 5, 7, 12, 13]
[25]	0.51	Fast and Accurate Time-Series Clustering	[2, 3, 4, 5, 8, 12]
[99]	0.46	Discovering similar multidimensional trajectories	[1, 4, 7, 12, 13]
[145]	0.45	A User-Friendly Guide to Using Distance Measures to Compare Time Series in Ecology	[1]
[146]	0.45	Benchmarking distance-based partitioning methods for mixed-type data	[1]
[111]	0.38	Fast subsequence matching in time-series databases	[2, 3, 4, 13]

Adjacent Work

Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

Ref.	Adjacency score	Title	References These Foundational Papers
[13]	0.12	k-Shape: Efficient and Accurate Clustering of Time Series	[22, 44, 51, 93, 99, 111, 134]
[57]	0.09	An empirical evaluation of similarity measures for time series classification	[22, 44, 93, 111, 134]
[4]	0.08	Debunking Four Long-Standing Misconceptions of Time-Series Distance Measures	[22, 44, 51, 93, 99, 111, 134]
[12]	0.08	A Survey on Time-Series Distance Measures	[4, 22, 44, 51, 93, 99]
[37]	0.07	Querying and mining of time series data: experimental comparison of representations and distance measures	[51, 93, 99, 111, 123]
[22]	0.07	Experimental comparison of representation methods and distance measures for time series data	[51, 93, 99, 111, 123]
[156]	0.06	Algorithm 1061: tsdistances: A High-Performance Python Library for Time Series Distances with GPU Support	[22, 37, 44, 51, 134]
[29]	0.06	Tensor Multi-Elastic Kernel Self-Paced Learning for Time Series Clustering	[22, 25, 51, 52, 134]
[157]	0.06	A New Lower Bounding Paradigm and Tighter Lower Bounds for Elastic Similarity Measures	[4, 37, 51, 93, 134]
[7]	0.06	Time-Series Clustering: A Comprehensive Study of Data Mining, Machine Learning, and Deep Learning Methods	[2, 3, 4, 22, 46, 93, 99, 123]
[3]	0.05	A Structured Study of Multivariate Time-Series Distance Measures	[4, 22, 51, 111, 134]
[158]	0.05	TSQ: An Optimized Framework for Efficiently Answering Time Series Queries	[37, 51, 93, 99, 111]
[99]	0.05	Discovering similar multidimensional trajectories	[111, 134]
[5]	0.05	A review and evaluation of elastic distance functions for time series clustering	[2, 46, 51, 93, 134]
[159]	0.04	HPS: A novel heuristic hierarchical pruning strategy for dynamic top-k trajectory similarity query	[37, 93, 99, 111, 113]
[160]	0.04	Spatiotemporal dynamics of human mobility after lifting of COVID-19 control measures in China: evidence from Nanjing City's Spring Festival	[37, 44, 134]
[161]	0.04	SPARTAN: Data-Adaptive Symbolic Time-Series Approximation	[3, 4, 5, 8, 12, 111]
[27]	0.04	Common Dissimilarity Measures are Inappropriate for Time Series Clustering	[37, 51, 56, 65, 93, 113, 126]

[74]	0.04	Accelerating Similarity Search for Elastic Measures: A Study and New Generalization of Lower Bounding Distances	[4, 51, 111, 113]
[125]	0.04	Time-Series Similarity and Clustering of Producer Share Dynamics in Agrifood Markets: Evidence from Origin–Destination Price Relationships	[44, 134]

References

- [1] [Comparing clustering approaches for smart meter time series: Investigating the influence of dataset properties on performance](#)
Luke W. Yerbury, ..., and Lachlan O'Neil. Applied Energy, 2024. 9 citations.
89% Topic Match
Presents a comprehensive empirical comparison of 31 distances and 11 clustering algorithms for smart meter time series.
Uses expert informed synthetic datasets emphasizing peak patterns, phased experiments (L1O selection, algorithm combos) and robustness tests varying cluster balance, noise, outliers.
Relevant because it directly compares ED, DTW, MSM, k sliding/SBD and related measures and studies noise and temporal shifts; however it does not appear to systematically vary sequence length nor evaluate distributional/kernel (IDK style) methods.
- [2] [A Benchmark Study on Time Series Clustering](#)
A. Javed, ..., and D. Rizzo. ArXiv, 2020. 141 citations.
63% Topic Match
Presents a large benchmark comparing ED, DTW, and SBD-like measures for time-series clustering.
Evaluates eight clustering algorithms on 112 equal-length UCR datasets, controlling six experimental factors and reporting per-dataset metrics.
Relevant as a broad, reproducible cross-paradigm benchmark (lock-step, elastic, sliding/shape) but omits MSM and distributional/kernel (IDK) methods and does not systematically vary noise level, warping amplitude, or sequence length.
- [3] [A Structured Study of Multivariate Time-Series Distance Measures](#)
Jens E. d'Hondt, ..., and John Paparrizos. Proceedings of the ACM on Management of Data, 2025. 10 citations.
51% Topic Match
Presents a comprehensive empirical evaluation of multivariate time-series distance measures.
Tests 30 measures across 8 categories (including lock-step, elastic, sliding, kernels), 2 channel-dependency models, 13 normalizations, 30 datasets, and 3 tasks with parameter tuning and statistical analysis.
Highly relevant: directly compares cross-paradigm distances (addresses prior gaps noted), examines normalization effects and statistical significance; less clear if it systematically varies noise level, controlled temporal misalignment, or sequence length as mechanistic factors.
- [4] [Debunking Four Long-Standing Misconceptions of Time-Series Distance Measures](#)
John Paparrizos, ..., and M. Franklin. Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data, 2020. 73 citations.
41% Topic Match
Demonstrates a comprehensive empirical evaluation of 71 time-series distance measures.
Tests 52 lock-step, 4 sliding, 7 elastic, 4 kernels, 4 embeddings across 128 datasets using 8 normalization schemes and rigorous statistical analyses.
Relevant as a broad, statistically careful cross-paradigm comparison, but it is primarily empirical/classification-focused (not mechanism-oriented); it does vary normalization and includes sliding/kernel families but does not systematically manipulate noise level, controlled temporal misalignment, or sequence length as independent experimental factors.
- [5] [A review and evaluation of elastic distance functions for time series clustering](#)
Christopher Holder, ..., and A. Bagnall. Knowledge and Information Systems, 2022. 67 citations.
38% Topic Match
Conducts a large empirical comparison of elastic distance measures for time-series clustering.
Implements nine elastic distances (DTW variants, ERP, EDR, LCSS, TWE, MSM, etc.), runs k-means and k-medoids on z-normalized UCR archive, reports accuracy/ARI and releases code.
Relevant as strong evidence that edit-war metrics (MSM, TWE) outperform pure DTW and that medoid-based clustering helps; NOT mechanistic for your needs—it only studies elastic paradigms on equal-length UCR data and does not systematically vary noise, temporal misalignment, sequence length, nor include SBD/shift-invariant or distributional/kernel methods.
- [6] [Efficiency of Time Series Clustering Method Based on Distribution of Difference Using Several Distances](#)
Phudit Thanakulkairid, ..., and Pat Vatiwutipong. 2022 19th International Joint Conference on Computer Science and Software Engineering (JCSSE), 2022. 0 citations.
32% Topic Match
Proposes clustering time series via distances between distributions of first-order differences.
Compares ED, KL divergence, Wasserstein, and DTW applied to empirical difference distributions; builds weighted graph and uses graph partitioning to cluster.
Relevant partially: explores a distributional view (difference distributions) and compares paradigms, but DTW is applied to distributions (not raw-sequence alignment), lacks systematic manipulation of noise/misalignment/length, and appears experimental-only with limited theoretical analysis.
- [7] [Time-Series Clustering: A Comprehensive Study of Data Mining, Machine Learning, and Deep Learning Methods](#)
John Paparrizos and Bogireddy Sai Prasanna Teja. Proc. VLDB Endow., 2025. 7 citations.
29% Topic Match
Evaluates and benchmarks 84 time-series clustering methods across 10 method classes.
Does so via a large empirical study on 128 datasets with rigorous statistical validation and fair comparison.
Limited mechanistic focus: compares many methods (including elastic/lock-step/deep models) but does not appear to systematically vary noise, temporal misalignment, or sequence length as controlled experimental factors; useful as broad empirical baseline but not a mechanism-oriented, cross-paradigm causal study.
- [8] [Odyssey: An Engine Enabling The Time-Series Clustering Journey](#)
John Paparrizos and Sai Prasanna Teja Reddy. Proc. VLDB Endow., 2023. 15 citations.
25% Topic Match
Presents a large-scale benchmarking engine for time-series clustering methods.
Implements and evaluates 80 clustering methods across 9 classes on 128 datasets with rigorous statistical analysis.
Useful as a broad empirical comparison (including elastic and other classes), but does not appear to systematically vary noise, controlled misalignment, or sequence length mechanistically.
- [9] [An Invariance-guided Stability Criterion for Time Series Clustering Validation](#)
Florent Forest, ..., and J. Lacaille. 2020 25th International Conference on Pattern Recognition (ICPR), 2021. 4 citations.
20% Topic Match
Proposes an invariance-guided clustering stability criterion for time-series model selection.
Implements stability analysis by perturbing series according to known invariances (shifts, warps, noise) and combining between-/within-cluster stability scores.
Relevant for mechanism-oriented comparisons because it explicitly models noise/misalignment perturbations for validation, but it does not empirically compare similarity paradigms (ED, DTW, MSM, SBD, IDK) nor analyze sequence-length effects.
- [10] [A user friendly guide to using distance measures to compare time series in ecology](#)
Shawn Dove, ..., and D. J. Murrell. Ecology and Evolution, 2023. 11 citations.
18% Topic Match

Provides a practical guide for selecting distance measures for time-series tasks.

Surveys 42 distances, tests them on 16 properties, and presents an objective selection workflow for ecology use.

Focus is applied/ecology-oriented (clustering mentioned) and empirical/property-driven; does not systematically compare elastic (DTW/MSM), SBD, and distributional/IDK paradigms across controlled noise, misalignment, or length regimes, so only somewhat related to your mechanism-oriented study.

[11] [Clustering time series by extremal dependence](#)

A. M. Alonso, ..., and M. Scotto. International Journal of Data Science and Analytics, 2024. 2 citations.

15% Topic Match

Abstract: The goal of this paper is to characterize the temporal dependence structure on the extremes of time series and use such dependency to group them. In particular, three similarity measures to capture extremal dependence are proposed, being their performance assessed in different scenarios. This will involve the use of classical time series clustering algorithms, as well as rigorous evaluation of their performance in both simulated scenarios and real-world time series data sets. The focus will be on comparing the performance of these similarity measures with different clustering methods, and illustrate the efficacy of extremal dependence-based clustering in meteorological data. To...

[12] [A Survey on Time-Series Distance Measures](#)

John Paparrizos, ..., and Odysseas Papapetrou. ArXiv, 2024. 14 citations.

13% Topic Match

Abstract: Distance measures have been recognized as one of the fundamental building blocks in time-series analysis tasks, e.g., querying, indexing, classification, clustering, anomaly detection, and similarity search. The vast proliferation of time-series data across a wide range of fields has increased the relevance of evaluating the effectiveness and efficiency of these distance measures. To provide a comprehensive view of this field, this work considers over 100 state-of-the-art distance measures, classified into 7 categories: lock-step measures, sliding measures, elastic measures, kernel measures, feature-based measures, model-based measures, and embedding measures. Beyond providing comprehensive mathematical frameworks, this work also delves into the distinctions and...

[13] [k-Shape: Efficient and Accurate Clustering of Time Series](#)

John Paparrizos and L. Gravano. SIGMOD Rec., 2016. 722 citations.

12% Topic Match

No summary or abstract available

[14] [Rock the KASBA: blazingly fast and accurate time series clustering](#)

Christopher Holder and Anthony J. Bagnall. Data Mining and Knowledge Discovery, 2024. 1 citations.

10% Topic Match

Abstract: Time series data has become increasingly prevalent across numerous domains, driving a growing demand for time series machine learning techniques. Among these, time series clustering (TSCL) stands out as one of the most popular machine learning tasks. TSCL serves as a powerful exploratory analysis tool and is also employed as a preprocessing step or subroutine for various tasks, including anomaly detection, segmentation, and classification. The most popular TSCL algorithms are either fast (in terms of runtime) but perform poorly on benchmark problems, or perform well on benchmarks but scale poorly. We present a new TSCL algorithm, the k-means (K) Accelerated...

[15] [Beyond the Dimensions: A Structured Evaluation of Multivariate Time Series Distance Measures](#)

Jens E. d'Hondt, ..., and John Paparrizos. 2024 IEEE 40th International Conference on Data Engineering Workshops (ICDEW), 2024. 8 citations.

9% Topic Match

Abstract: A variety of distance measures for multivariate time series has been proposed in recent literature. However, evaluations of such measures have been incomplete; comparisons are limited to subsets of similar measures, lacking a holistic view of the field with an appropriate taxonomy of measures. This paper presents a structured evaluation of multivariate time series distance measures. Through a novel taxonomy, measures are categorized based on how they handle the multiple variates; in an atomic or a holistic manner. Experimental evaluation of 12 measures shows that no single measure or approach is superior; the optimal choice depends on the data and...

[16] [Soft-DTW: a Differentiable Loss Function for Time-Series](#)

Marco Cuturi and Mathieu Blondel. Unknown journal, 2017. 806 citations.

9% Topic Match

Abstract: We propose in this paper a differentiable learning loss between time series, building upon the celebrated dynamic time warping (DTW) discrepancy. Unlike the Euclidean distance, DTW can compare time series of variable size and is robust to shifts or dilatations across the time dimension. To compute DTW, one typically solves a minimal-cost alignment problem between two time series using dynamic programming. Our work takes advantage of a smoothed formulation of DTW, called soft-DTW, that computes the soft-minimum of all alignment costs. We show in this paper that soft-DTW is a differentiable loss function, and that both its value and gradient...

[17] [Comparison of Model-Free Algorithms For Clustering GARCH Processes](#)

G. Adamyan. Mathematical Problems of Computer Science, 2022. 0 citations.

8% Topic Match

Abstract: In this paper, we evaluate several model-free algorithms for clustering time series datasets generated by GARCH processes. In extensive experiments, we generate synthetic datasets in different scenarios. Then, we compare K-Means (for Euclidian and dynamic time warping distance), K-Shape, and Kernel K-Means models with different clustering metrics. Several experiments show that the K-Means model with dynamic time warping distance archives comparably better results. However, the considered models have significant shortcomings in improving the clustering accuracy when the amount of information (the minimum length of the time series) increases, and in performing accurate clustering when data is unbalanced or clusters are...

[18] [Distribution-Based Trajectory Clustering](#)

Zijing Wang, ..., and Kai Ming Ting. 2023 IEEE International Conference on Data Mining (ICDM), 2023. 1 citations.

7% Topic Match

Abstract: Trajectory clustering enables the discovery of common patterns in trajectory data. Current methods of trajectory clustering rely on a distance measure between two points in order to measure the dissimilarity between two trajectories, causing problems of both effectiveness and efficiency. In this paper, we propose a new IDK-based clustering algorithm, called TIDKC, which makes full use of the distributional kernel for trajectory similarity measuring and clustering. TIDKC identifies non-linearly separable clusters with irregular shapes and varied densities in linear time. It does not rely on random initialisation and is robust to outliers. An extensive evaluation on 7 large real-world trajectory...

[19] [Time-Series Clustering Based on the Characterization of Segment Typologies](#)

David Guijo-Rubio, ..., and C. Hervás Martínez IEEE Transactions on Cybernetics, 2018. 51 citations.

6% Topic Match

Abstract: Time-series clustering is the process of grouping time series with respect to their similarity or characteristics. Previous approaches usually combine a specific distance measure for time series and a standard clustering method. However, these approaches do not take the similarity of the different subsequences of each time series into account, which can be used to better compare the time-series objects of the dataset. In this article, we propose a novel technique of time-series clustering consisting of two clustering stages. In a first step, a least-squares polynomial segmentation procedure is applied to each time series, which is based on a growing...

[20] [Time-Series Clustering in R Using the dtwclust Package](#)

Alexis Sarda-Espinosa. R J., 2019. 144 citations.

6% Topic Match

Abstract: Most clustering strategies have not changed considerably since their initial definition. The common improvements are either related to the distance measure used to assess dissimilarity, or the function used to calculate prototypes. Time-series clustering is no exception, with the Dynamic Time Warping distance being particularly popular in that context. This distance is computationally expensive, so many related optimizations have been developed over the years. Since no single clustering algorithm can be said to perform best on all datasets, different strategies must be tested and compared, so a common infrastructure can be advantageous. In this manuscript, a general overview of shape-based...

[21] [From Time Series to Euclidean Spaces: On Spatial Transformations for Temporal Clustering](#)

Nuno Gonçalves, ..., and Anika Schumann. ArXiv, 2020. 0 citations.

5% Topic Match

Abstract: Unsupervised clustering of temporal data is both challenging and crucial in machine learning. In this paper, we show that neither traditional clustering methods, time series specific or even deep learning-based alternatives generalise well when both varying sampling rates and high dimensionality are present in the input data. We propose a novel approach to temporal clustering, in which we (1) transform the input time series into a distance-based projected representation by using similarity measures suitable for dealing with temporal data,(2) feed these projections into a multi-layer CNN-GRU autoencoder to generate meaningful domain-aware latent representations, which ultimately (3) allow for a natural...

- [22] [Experimental comparison of representation methods and distance measures for time series data](#)

Xiaoyue Wang, ..., and Eamonn J. Keogh. Data Mining and Knowledge Discovery, 2010. 883 citations.

5% Topic Match

Abstract: The previous decade has brought a remarkable increase of the interest in applications that deal with querying and mining of time series data. Many of the research efforts in this context have focused on introducing new representation methods for dimensionality reduction or novel similarity measures for the underlying data. In the vast majority of cases, each individual work introducing a particular method has made specific claims and, aside from the occasional theoretical justifications, provided quantitative experimental observations. However, for the most part, the comparative aspects of these experiments were too narrowly focused on demonstrating the benefits of the proposed methods...

- [23] [Elastic similarity and distance measures for multivariate time series](#)

Ahmed Shifaz, ..., and Geoffrey I. Webb. Knowledge and Information Systems, 2021. 31 citations.

4% Topic Match

Abstract: This paper contributes multivariate versions of seven commonly used elastic similarity and distance measures for time series data analytics. Elastic similarity and distance measures can compensate for misalignments in the time axis of time series data. We adapt two existing strategies used in a multivariate version of the well-known Dynamic Time Warping (DTW), namely, Independent and Dependent DTW, to these seven measures. While these measures can be applied to various time series analysis tasks, we demonstrate their utility on multivariate time series classification using the nearest neighbor classifier. On 23 well-known datasets, we demonstrate that each of the measures but...

- [24] [Multi-Metric Fusion-Based Ensemble Clustering for Time Series via Diverse Elastic Distance Functions](#)

Baohan Mi, ..., and Jinming Du. 2025 IEEE 23rd International Conference on Industrial Informatics (INDIN), 2025. 0 citations.

4% Topic Match

Abstract: The growing volume of unlabeled industrial process data necessitates the development of advanced time series clustering (TSCL) algorithms for exploratory analysis. To overcome the inherent limitations of conventional single-clustering algorithms in handling complex temporal patterns and enhance robustness, a novel ensemble TSCL framework, integrating multi-metric fusion and diverse elastic distance functions, is proposed in this article. Initially, to extract multi-domain patterns, diverse base clusters are fit for augmented time series sequences via diverse elastic distance measures, with the performance evaluated on the training subsets through multiple internal clustering validation metrics. Then, a multi-metric fusion scheme, leveraging entropy to balance the...

- [25] [Fast and Accurate Time-Series Clustering](#)

John Paparrizos and L. Gravano. ACM Transactions on Database Systems (TODS), 2017. 205 citations.

4% Topic Match

No summary or abstract available

- [26] [Adaptive Clustering Framework for Time-Series Data Using Enhanced FastDTW](#)

Jiahao Tao, ..., and Chen Liang. 2025 37th Chinese Control and Decision Conference (CCDC), 2025. 0 citations.

3% Topic Match

Abstract: Clustering time-series data presents considerable challenges, driven by its inherent high dimensionality, dynamic variability, and nonlinear temporal dependencies. To overcome these obstacles, this study introduces an Adaptive Clustering Framework for Time-Series Data, built upon the foundation of Adaptive Constraint FastDTW (ACFastDTW) and further extended with the enhanced ACFastDTW-SALKM. The ACFastDTW algorithm utilizes a CatBoost-regressed adaptive windowing mechanism, which significantly boosts both the accuracy of distance calculations and computational efficiency. Building on this foundation, ACFastDTW-SALKM employs a self-attention LSTM autoencoder for extracting critical features and combines it with K-means clustering to achieve precise segmentation of time-series data. Through validation on both...

- [27] [Common Dissimilarity Measures are Inappropriate for Time Series Clustering](#)

Cássio M. M. Pereira and R. Mello. RITA, 2013. 8 citations.

3% Topic Match

Abstract: Clustering algorithms have been actively used to identify similar timeseries, providing a better understanding of data. However, common clustering dissimilarity measures disregard time series correlations, yielding poor results. In thispaper, we introduce a dissimilarity measure based on series partial autocorrelations.Experiments compare hierarchical clustering algorithms using the common dissimilarity measures, such as Euclidean Distance and Dynamic Time Warping, to clustertime series following Box-Jenkins Auto-Regressive models. Results show that ourdissimilarity measure produces better results for both synthetic and real data sets interms of the Adjusted Rand Index and Normalized Hubert *statistic. Our findingsconfirm that the choice of dissimilarity measure is crucial...

- [28] [On time series clustering with k-means](#)

Christopher Holder, ..., and Jason Lines. ArXiv, 2024. 3 citations.

3% Topic Match

Abstract: There is a long history of research into time series clustering using distance-based partitional clustering. Many of the most popular algorithms adapt k-means (also known as Lloyd's algorithm) to exploit time dependencies in the data by specifying a time series distance function. However, these algorithms are often presented with k-means configured in various ways, altering key parameters such as the initialisation strategy. This variability makes it difficult to compare studies because k-means is known to be highly sensitive to its configuration. To address this, we propose a standard Lloyd's-based model for TSCL that adopts an end-to-end approach, incorporating a specialised...

- [29] [Tensor Multi-Elastic Kernel Self-Paced Learning for Time Series Clustering](#)

Yongqiang Tang, ..., and Wensheng Zhang. IEEE Transactions on Knowledge and Data Engineering, 2019. 36 citations.

2% Topic Match

Abstract: Time series clustering has attracted growing attention due to the abundant data accessible and extensive value in various applications. The unique characteristics of time series, including high-dimension, warping, and the integration of multiple elastic measures, pose challenges for the present clustering algorithms, most of which take into account only part of these difficulties. In this paper, we make an effort to simultaneously address all aforementioned issues in time series clustering under a unified multiple kernels clustering (MKC) framework. Specifically, we first implicitly map the raw time series space into multiple kernel spaces via elastic distance measure functions. In such high-dimensional...

- [30] [Clustering Noisy Time Series](#)

Anastasiiia Tkachenko, ..., and Tamara Anatoliivna Radyvylova. System technologies, 2019. 0 citations.

2% Topic Match

Abstract: One of the urgent tasks of machine learning is the problem of clustering objects. Clustering time series is used as an independent research technique, as well as part of more complex data mining methods, such as rule detection, classification, anomaly detection, etc.A comparative analysis of clustering noisy time series is carried out. The clustering sample contained time series of various types, among which there were atypical objects. Clustering was performed by k-means and DBSCAN methods using various distance functions for time series.A numerical experiment was conducted to investigate the application of the k-means and DBSCAN methods to model time series...

- [31] [Equivalence partition based morphological similarity clustering for large-scale time series](#)

Shaolin Hu. Scientific Reports, 2023. 3 citations.

2% Topic Match

Abstract: Data clustering belongs to the category of unsupervised learning and plays an important role in the dynamic systems and big data. The clustering problem of sampled time-series data is undoubtedly much more challenging than that of repeatable sampling data. Most of the existing time-series clustering methods stay at the level of algorithm design, lacking rigorous theoretical foundation and being inefficient in dealing with large-scale time series. To address this issue, in this paper, we establish the mathematical theory for the large-scale time series clustering of dynamic system. The main contributions of this paper include proposing the concept of time series...

[32] [The Performance of Distances Between Time Series: An In Depth Comparison](#)

Margarida G. M. S. Cardoso and Ana Martins. Expert Systems, 2025. 0 citations.

2% Topic Match

Abstract: The performance of distance measures between time series has been discussed in diverse studies. Most identified performance as the accuracy resulting from the use of a specific distance in 1 Nearest Neighbour. Few studies have addressed the related computation time, and no systematic analyses of the associations between the distances' performance (1 NN based accuracy and computation time) and the time series' characteristics have been presented yet. We propose to fill this research gap by analysing these relationships considering the following features: the training and test sets' dimensions, the time series' length, the number of classes, and the classes' separability as measured by...

[33] [In-Database Time Series Clustering](#)

Yunxiang Su, ..., and Shaoxu Song. Proceedings of the ACM on Management of Data, 2025. 2 citations.

2% Topic Match

Abstract: Time series data are often clustered repeatedly across various time ranges to mine frequent subsequence patterns from different periods, which could further support downstream applications. Existing state-of-the-art (SOTA) time series clustering method, such as K-Shape, can proficiently cluster time series data referring to their shapes. However, in-database time series clustering problem has been neglected, especially in IoT scenarios with large-volume data and high efficiency demands. Most time series databases employ LSM-Tree based storage to support intensive writings, yet causing underlying data points out-of-order in timestamps. Therefore, to apply existing out-of-database methods, all data points must be fully loaded into memory...

[34] [Time Series Cluster Kernel for Learning Similarities between Multivariate Time Series with Missing Data](#)

Karl Øyvind Mikalsen, ..., and R. Jenssen. ArXiv, 2017. 99 citations.

1% Topic Match

Abstract: Abstract Similarity-based approaches represent a promising direction for time series analysis. However, many such methods rely on parameter tuning, and some have shortcomings if the time series are multivariate (MTS), due to dependencies between attributes, or the time series contain missing data. In this paper, we address these challenges within the powerful context of kernel methods by proposing the robust time series cluster kernel (TCK). The approach taken leverages the missing data handling properties of Gaussian mixture models (GMM) augmented with informative prior distributions. An ensemble learning approach is exploited to ensure robustness to parameters by combining the clustering results...

[35] [Robust DTW-based entropy fuzzy clustering of time series](#)

P. D'Urso, ..., and V. Vitale. Annals of Operations Research, 2023. 14 citations.

1% Topic Match

Abstract: Time series are complex data objects whose partitioning into homogeneous groups is still a challenging task, especially in the presence of outliers or noisy data. To address the problem of robustness against outliers in clustering techniques, this paper proposes a robust fuzzy C-medoids method based on entropy regularization. In-depth, we use an appropriate exponential transformation of the dissimilarity based on Dynamic Time Warping, which can be computed also for time series of different length. In addition, the fuzzy framework provides the necessary flexibility to cope with the complexity of the features space. It allows a time series to be assigned...

[36] [A Comparison Study on Similarity and Dissimilarity Measures in Clustering Continuous Data](#)

A. S. Shirkhorshidi, ..., and A. Dalby. PLoS ONE, 2015. 374 citations.

1% Topic Match

Abstract: Similarity or distance measures are core components used by distance-based clustering algorithms to cluster similar data points into the same clusters, while dissimilar or distant data points are placed into different clusters. The performance of similarity measures is mostly addressed in two or three-dimensional spaces, beyond which, to the best of our knowledge, there is no empirical study that has revealed the behavior of similarity measures when dealing with high-dimensional datasets. To fill this gap, a technical framework is proposed in this study to analyze, compare and benchmark the influence of different similarity measures on the results of distance-based clustering...

[37] [Querying and mining of time series data: experimental comparison of representations and distance measures](#)

Hui Ding, ..., and Eamonn J. Keogh. Proc. VLDB Endow., 2008. 1481 citations.

1% Topic Match

No summary or abstract available

[38] [Clustering Financial Time Series](#)

Nshan Potikyan, ..., and Assoc. Prof Tartu. Journal Not Provided, Unknown year. 0 citations.

1% Topic Match

Abstract: Time series clustering is heavily based on choosing a proper dissimilarity measure between a pair of time series. We present several dissimilarity measures and use two synthetic datasets to evaluate their performance. Hierarchical clustering and network analysis methods are used to perform cluster analysis on stock price time series of 594 US-based companies in order to verify whether stock prices of companies operating within an industry have common fluctuations. The results of the thesis show that some companies within the same industry do form clusters, while others are relatively scattered.

[39] [A Hybrid Algorithm for Clustering of Time Series Data Based on Affinity Search Technique](#)

S. Aghabozorgi, ..., and Alireza Jalali. The Scientific World Journal, 2014. 58 citations.

1% Topic Match

Abstract: Time series clustering is an important solution to various problems in numerous fields of research, including business, medical science, and finance. However, conventional clustering algorithms are not practical for time series data because they are essentially designed for static data. This impracticality results in poor clustering accuracy in several systems. In this paper, a new hybrid clustering algorithm is proposed based on the similarity in shape of time series data. Time series data are first grouped as subclusters based on similarity in time. The subclusters are then merged using the k-Medoids algorithm based on similarity in shape. This model has...

[40] [Ranked differences Pearson correlation dissimilarity with an application to electricity users time series clustering](#)

Chutipan Charoensuk and Nathakhun Wiroonsri. ArXiv, 2025. 2 citations.

1% Topic Match

Abstract: Time series clustering is an unsupervised learning method for classifying time series data into groups with similar behavior. It is used in applications such as healthcare, finance, economics, energy, and climate science. Several time series clustering methods have been introduced and used for over four decades. Most of them focus on measuring either Euclidean distances or association dissimilarities between time series. In this work, we propose a new dissimilarity measure called ranked Pearson correlation dissimilarity (RDPC), which combines a weighted average of a specified fraction of the largest element-wise differences with the well-known Pearson correlation dissimilarity. It is incorporated into...

[41] [The UCR time series archive](#)

Hoang Anh Dau, ..., and Eamonn J. Keogh. IEEE/CAA Journal of Automatica Sinica, 2018. 1041 citations.

1% Topic Match

Abstract: The UCR time series archive—introduced in 2002, has become an important resource in the time series data mining community, with at least one thousand published papers making use of at least one data set from the archive. The original incarnation of the archive had sixteen data sets but since that time, it has gone through periodic expansions. The last expansion took place in the summer of 2015 when the archive grew from 45 to 85 data sets. This paper introduces and will focus on the new data expansion from 85 to 128 data sets. Beyond expanding this valuable resource, this...

[42] [Quantile autocovariances: A powerful tool for hard and soft partitional clustering of time series](#)

J. A. Vilar, ..., and P. D'Urso. Fuzzy Sets Syst., 2017. 44 citations.

1% Topic Match

Abstract: A key issue in cluster analysis is determining a proper dissimilarity measure between two data objects, and many pairwise dissimilarities have been proposed to deal with time series. Assuming that the clustering purpose is to group series according to the underlying dependence structures, a detailed study of the behavior in clustering of a dissimilarity based on comparing estimated quantile autocovariance functions (QAF) is carried out. Quantile autocovariances provide information about the serial dependence structure that other conventional features are not able to capture, which suggests great potential to perform clustering of series. The asymptotic behavior of the sample quantile...

[43] [ShiftDTW: adapting the DTW metric for cyclic time series clustering](#)

Lucas Foulon, ..., and Xavier Millot. ArXiv, 2024. 0 citations.

1% Topic Match

Abstract: The elasticity of the DTW metric provides a more flexible comparison between time series and is used in numerous machine learning domains such as classification or clustering. However, it does not align the measurements at the beginning and end of time series if they have a shift occurring right at the start of one series, with the omitted part appearing at the end of that series. Due to the cyclicity of such series - which lack a definite beginning or end - we rely on the Cyclic DTW approach to propose a less computationally expensive approximation of this calculation method....

[44] [Clustering of time series data - a survey](#)

T. Liao. Pattern Recognit., 2005. 2627 citations.

1% Topic Match

No summary or abstract available

[45] [Time Series Clustering with Topological and Geometric Mixed Distance](#)

Yunsheng Zhang, ..., and Haifeng Li. Mathematics, 2021. 9 citations.

1% Topic Match

Abstract: Time series clustering is an essential ingredient of unsupervised learning techniques. It provides an understanding of the intrinsic properties of data upon exploiting similarity measures. Traditional similarity-based methods usually consider local geometric properties of raw time series or the global topological properties of time series in the phase space. In order to overcome their limitations, we put forward a time series clustering framework, referred to as time series clustering with Topological-Geometric Mixed Distance (TGMD), which jointly considers local geometric features and global topological characteristics of time series data. More specifically, persistent homology is employed to extract topological features of time...

[46] [The Move-Split-Merge Metric for Time Series](#)

Alexandra Stefan, ..., and Gautam Das. IEEE Transactions on Knowledge and Data Engineering, 2013. 195 citations.

0% Topic Match

No summary or abstract available

[47] [Spectral-adaptive masking and hierarchical contrastive learning for time series clustering](#)

Zhixuan Wang, ..., and Yuyang Shen. Intelligent Data Analysis: An International Journal, 2026. 0 citations.

0% Topic Match

Abstract: Time series clustering is a pivotal technique for efficiently mining the structure of data. However, time series data possess unique characteristics such as periodicity, nonlinearity and sensitivity to high-frequency noise in dynamic environments, which significantly impact the performance of clustering methods. Recently, deep clustering has garnered widespread attention for its outstanding performance in capture multi-scale temporal dependencies. Despite this, existing methods struggle to effectively captured the similarities and diverse temporal patterns in noisy time series. Accordingly, we propose a novel deep time series clustering framework integrating spectral-adaptive masking with hierarchical contrastive learning. First, an enhanced encoder is designed to generate...

[48] [The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances](#)

A. Bagnall, ..., and Eamonn J. Keogh. Data Mining and Knowledge Discovery, 2016. 1429 citations.

0% Topic Match

Abstract: In the last 5 years there have been a large number of new time series classification algorithms proposed in the literature. These algorithms have been evaluated on subsets of the 47 data sets in the University of California, Riverside time series classification archive. The archive has recently been expanded to 85 data sets, over half of which have been donated by researchers at the University of East Anglia. Aspects of previous evaluations have made comparisons between algorithms difficult. For example, several different programming languages have been used, experiments involved a single train/test split and some used normalised data whilst others...

[49] [CSTS: A Benchmark for the Discovery of Correlation Structures in Time Series Clustering](#)

Isabella Degen, ..., and K. Brown. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Time series clustering promises to uncover hidden structural patterns in data with applications across healthcare, finance, industrial systems, and other critical domains. However, without validated ground truth information, researchers cannot objectively assess clustering quality or determine whether poor results stem from absent structures in the data, algorithmic limitations, or inappropriate validation methods, raising the question whether clustering is "more art than science" (Guyon et al., 2009). To address these challenges, we introduce CSTS (Correlation Structures in Time Series), a synthetic benchmark for evaluating the discovery of correlation structures in multivariate time series data. CSTS provides a clean benchmark that enables researchers to...

[50] [Matrix Profile XII: MPdist: A Novel Time Series Distance Measure to Allow Data Mining in More Challenging Scenarios](#)

Shaghayegh Gharghabi, ..., and Eamonn J. Keogh. 2018 IEEE International Conference on Data Mining (ICDM), 2018. 54 citations.

0% Topic Match

Abstract: At their core, many time series data mining algorithms can be reduced to reasoning about the shapes of time series subsequences. This requires a distance measure, and most algorithms use Euclidean Distance or Dynamic Time Warping (DTW) as their core subroutine. We argue that these distance measures are not as robust as the community believes. The undue faith in these measures derives from an overreliance on benchmark datasets and self-selection bias. The community is reluctant to address more difficult domains, for which current distance measures are ill-suited. In this work, we introduce a novel distance measure MPdist. We show that...

[51] [On The Marriage of Lp-norms and Edit Distance](#)

Lei Chen and R. Ng. Unknown journal, 2004. 852 citations.

0% Topic Match

No summary or abstract available

[52] [Time Warp Edit Distance with Stiffness Adjustment for Time Series Matching](#)

P. Marteau. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2007. 356 citations.

0% Topic Match

Abstract: In a way similar to the string-to-string correction problem, we address discrete time series similarity in light of a time-series-to-time-series-correction problem for which the similarity between two time series is measured as the minimum cost sequence of edit operations needed to transform one time series into another. To define the edit operations, we use the paradigm of a graphical editing process and end up with a dynamic programming algorithm that we call time warp edit distance (TWED). TWED is slightly different in form from dynamic time warping (DTW), longest common subsequence (LCSS), or edit distance with real penalty (ERP) algorithms...

[53] [DTW-Distance Based Kernel for Time Series Data](#)

Hiroyuki Narita, ..., and A. Hayashi. IEICE Trans. Inf. Syst., 2009. 3 citations.

0% Topic Match

Abstract: One of the advantages of the kernel methods is that they can deal with various kinds of objects, not necessarily vectorial data with a fixed number of attributes. In this paper, we develop kernels for time series data using dynamic time warping (DTW) distances. Since DTW distances are pseudo distances that do not satisfy the triangle inequality, a kernel matrix based on them is not positive semidefinite, in general. We use semidefinite programming (SDP) to guarantee the positive definiteness of a kernel matrix. We present neighborhood preserving embedding (NPE), an SDP formulation to obtain a kernel matrix that best preserves...

[54] [Clustering-enhanced stock price prediction using deep learning](#)

Man Li, ..., and M. Angelova. World Wide Web, 2022. 51 citations.

0% Topic Match

Abstract: In recent years, artificial intelligence technologies have been successfully applied in time series prediction and analytic tasks. At the same time, a lot of attention has been paid to financial time series prediction, which targets the development of novel deep learning models or optimize the forecasting results. To optimize the accuracy of stock price prediction, in this paper, we propose a clustering-enhanced deep learning framework to predict stock prices with three matured deep learning forecasting models, such as Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN) and Gated Recurrent Unit (GRU). The proposed framework considers the clustering as the forecasting...

[55] [Multiscale Dubuc: A New Similarity Measure for Time Series](#)

Mahsa Khazaei, ..., and Krishna Rukmini Puthucode. 2024 IEEE International Conference on Big Data (BigData), 2024. 2 citations.

0% Topic Match

Abstract: Quantifying similarities between time series in a meaningful way remains a challenge in time series analysis, despite many advances in the field. Most real-world solutions still rely on a few popular measures, such as Euclidean Distance (EuD), Longest Common Subsequence (LCSS), and Dynamic Time Warping (DTW). The strengths and weaknesses of these measures have been studied extensively, and incremental improvements have been proposed. In this study, however, we present a different similarity measure that fuses the notion of Dubuc's variation from fractal analysis with the Intersection- over-Union (IoU) measure which is widely used in object recognition (also known as the...

[56] [Searching and Mining Trillions of Time Series Subsequences under Dynamic Time Warping](#)

T. Rakthanmanon, ..., and Eamonn J. Keogh. KDD : proceedings. International Conference on Knowledge Discovery & Data Mining, 2012. 1102 citations.

0% Topic Match

Abstract: Most time series data mining algorithms use similarity search as a core subroutine, and thus the time taken for similarity search is the bottleneck for virtually all time series data mining algorithms. The difficulty of scaling search to large datasets largely explains why most academic work on time series data mining has plateaued at considering a few millions of time series objects, while much of industry and science sits on billions of time series objects waiting to be explored. In this work we show that by using a combination of four novel ideas we can search and mine truly massive...

[57] [An empirical evaluation of similarity measures for time series classification](#)

J. Serrà and J. Arcos. ArXiv, 2014. 238 citations.

0% Topic Match

Abstract: Time series are ubiquitous, and a measure to assess their similarity is a core part of many computational systems. In particular, the similarity measure is the most essential ingredient of time series clustering and classification systems. Because of this importance, countless approaches to estimate time series similarity have been proposed. However, there is a lack of comparative studies using empirical, rigorous, quantitative, and large-scale assessment strategies. In this article, we provide an extensive evaluation of similarity measures for time series classification following the aforementioned principles. We consider 7 different measures coming from alternative measure 'families', and 45 publicly-available time series...

[58] [Feature-driven Time Series Clustering](#)

D. Tiano, ..., and Raymond Ng. Unknown journal, 2021. 9 citations.

0% Topic Match

Abstract: The problem of clustering time series has several applications in real-life contexts, especially in data science and data analytics pipelines. Existing time series clustering algorithms are ineffective for feature-rich real-world time series since they only compute the similarity of time series based on raw data or use a fixed set of features. In this paper, we develop a feature-based semi-supervised clustering framework addressing the above issues for variable-length and heterogeneous time series. Specifically, we rely on a graph encoding of the time series that is obtained by considering a high number of significant extracted features. We then employ community detection...

[59] [Accelerating Dynamic Time Warping Clustering with a Novel Admissible Pruning Strategy](#)

Nurjahan Begum, ..., and Eamonn J. Keogh. Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2015. 134 citations.

0% Topic Match

No summary or abstract available

[60] [Scalable Machine Learning: Development of Efficient Clustering Algorithms for High-Volume Temporal Data Analysis](#)

Madhur Taneja. 2023 International Conference on Communication, Security and Artificial Intelligence (ICCSAI), 2023. 1 citations.

0% Topic Match

Abstract: In the era of big data, the analysis of large-scale time series data has gained significant importance across various applications such as finance and healthcare. However, traditional clustering approaches, often encounter accuracy and scalability issues when applied to extensive data sets. This paper introduces innovative clustering method specifically tailored for time series data to enter these challenges. These methods leverage advanced data structures for computational efficiency, optimal distance measurements to improve, improve accuracy, and parallel computing framework for enhanced Scalability. Rig testing involving millions of time series occurrences from real world. Data sets was conducted to evaluate the effectiveness of...

[61] [Visual Extraction of Interaction Patterns Guided by Hierarchical Clustering and Process Mining](#)

Peilin Yu, ..., and K. Vrotsou. IEEE Transactions on Visualization and Computer Graphics, 2025. 0 citations.

0% Topic Match

Abstract: Understanding user interactions in digital systems is essential in analyzing user behaviors and improving system usability. However, a collection of interaction sequences is often large and unstructured, making it challenging to uncover interaction patterns. To address this challenge, we introduce a visual analytics approach that integrates hierarchical clustering and process mining techniques to support analysts in exploring unstructured, large interaction sequence data. Our system employs a tailored dynamic time warping-based similarity measure to enable comparison of interaction sequences. Based on the sequence similarities, we provide stepwise, interactive navigation of clustering results with contextual visual cues for refinement and validation. We...

[62] [The Influence of Global Constraints on Similarity Measures for Time-Series Databases](#)

V. Kurbalija, ..., and M. Ivanovi. ArXiv, 2011. 49 citations.

0% Topic Match

Abstract: A time series consists of a series of values or events obtained over repeated measurements in time. Analysis of time series represents an important tool in many application areas, such as stock-market analysis, process and quality control, observation of natural phenomena, and medical diagnosis. A vital component in many types of time-series analyses is the choice of an appropriate distance/similarity measure. Numerous measures have been proposed to date, with the most successful ones based on dynamic programming. Being of quadratic time complexity, however, global constraints are often employed to limit the search space in the matrix during the dynamic programming...

[63] [A New Distributional Treatment for Time Series and An Anomaly Detection Investigation](#)

K. Ting, ..., and Ye Zhu. Proc. VLDB Endow., 2022. 15 citations.

0% Topic Match

Abstract: Time series is traditionally treated with two main approaches, i.e., the time domain approach and the frequency domain approach. These approaches must rely on a sliding window so that time-shift versions of a periodic subsequence can be measured to be similar. Coupled with the use of a root point-to-point measure,

existing methods often have quadratic time complexity. We offer the third R domain approach. It begins with an insight that subsequences in a periodic time series can be treated as sets of independent and identically distributed (iid) points generated from an unknown distribution in R. This R domain treatment enables...

[64] [Evaluating DTW Measures via a Synthesis Framework for Time-Series Data](#)

Kishansingh Rajput, ..., and Guoning Chen. ArXiv, 2024. 4 citations.

0% Topic Match

Abstract: Time-series data originate from various applications that describe specific observations or quantities of interest over time. Their analysis often involves the comparison across different time-series data sequences, which in turn requires the alignment of these sequences. Dynamic Time Warping (DTW) is the standard approach to achieve an optimal alignment between two temporal signals. Different variations of DTW have been proposed to address various needs for signal alignment or classifications. However, a comprehensive evaluation of their performance in these time-series data processing tasks is lacking. Most DTW measures perform well on certain types of time-series data without a clear explanation of...

[65] [An efficient and accurate method for evaluating time series similarity](#)

Michael D. Morse and J. Patel. Unknown journal, 2007. 200 citations.

0% Topic Match

No summary or abstract available

[66] [Affine and Regional Dynamic Time Warping](#)

Tsu-Wei Chen, ..., and D. Stashuk. 2015 IEEE International Conference on Data Mining Workshop (ICDMW), 2015. 8 citations.

0% Topic Match

Abstract: Pointwise matches between two time series are of great importance in time series analysis, and dynamic time warping (DTW) is known to provide generally reasonable matches. There are situations where time series alignment should be invariant to scaling and offset in amplitude or where local regions of the considered time series should be strongly reflected in pointwise matches. Two different variants of DTW, affine DTW (ADTW) and regional DTW (RDTW), are proposed to handle scaling and offset in amplitude and provide regional emphasis respectively. Furthermore, ADTW and RDTW can be combined in two different ways to generate alignments that incorporate...

[67] [Gait pattern stratification in persons with incomplete spinal cord injury: a data-driven approach](#)

M. Truong, ..., and E. Gutierrez-Farewik. Journal of NeuroEngineering and Rehabilitation, 2026. 0 citations.

0% Topic Match

Abstract: Background Incomplete spinal cord injury (ISCI) often causes heterogeneous locomotion dysfunctions, depending on remaining sensorimotor function. Clinical tests and traditional gait analysis have limited ability to quantify the diversity of gait impairments. Unsupervised learning techniques can objectively identify common gait patterns among the overall heterogeneity. Explainable artificial intelligence approaches, when combined with machine learning models, can reveal important features often missed by traditional gait analyses. This study presents a framework to characterize gait heterogeneity among persons with ISCI based on several data-driven methods. We aimed to stratify overall gait heterogeneity by deriving clusters with similarities without a priori identification of...

[68] [Clustering and Classification for Time Series Data in Visual Analytics: A Survey](#)

Mohammed Ali, ..., and Xianghua Xie. IEEE Access, 2019. 72 citations.

0% Topic Match

Abstract: Visual analytics for time series data has received a considerable amount of attention. Different approaches have been developed to understand the characteristics of the data and obtain meaningful statistics in order to explore the underlying processes, identify and estimate trends, make decisions and predict the future. The machine learning and visualization areas share a focus on extracting information from data. In this paper, we consider not only automatic methods but also interactive exploration. The ability to embed efficient machine learning techniques (clustering and classification) in interactive visualization systems is highly desirable in order to gain the most from both humans...

[69] [Fast Constrained Dynamic Time Warping for Similarity Measure of Time Series Data](#)

Wonyoung Choi, ..., and Yunho Jung. IEEE Access, 2020. 53 citations.

0% Topic Match

Abstract: In this paper, we propose an efficient algorithm for reducing the computational complexity of dynamic time warping (DTW) for obtaining similarity measures between time series. The DTW technique exhibits superior classification accuracy compared to other algorithms but has a limitation of high computational complexity. To reduce the computational complexity of standard DTW, constrained DTW and fast DTW techniques have been proposed. The constrained DTW technique reduces the computational complexity of standard DTW by only considering limited alignments and prevent excessive alignments between two time series, which can reduce the overall classification accuracy. However, since the searching window for limited alignments...

[70] [Quantifying morphology changes in time series data with skew](#)

Phil Sung, ..., and J. Gutttag. 2009 IEEE International Conference on Acoustics, Speech and Signal Processing, 2009. 11 citations.

0% Topic Match

Abstract: This paper examines strategies to quantify differences in the morphology of time series while accounting for time skew in the observed data. We adapt four measures originally designed for signal shape comparison: Dynamic Time-Warping (DTW), Earth Mover's Distance (EMD), Fréchet Distance (FD), and Hausdorff Distance (HD). These morphology difference metrics on time series are compared in discriminative power and noise resistance on ECG signals as well as on a synthetic dataset. We use data from our experiments to shed light on the relative strengths of the methods.

[71] [Agrupamento de séries temporais em fluxos contínuos de dados](#)

C. M. M. Pereira. Unknown journal, 2013. 2 citations.

0% Topic Match

Abstract: Recently, the data streams mining area has gained importance, which aims to extract useful information from massive and continuous data sources that evolve over time. One of the most popular techniques in this area is clustering, which aims to structure large volumes of data into hierarchies or partitions, such that similar objects are placed in the same group. Several algorithms were proposed in this context, however most of them focused on the clustering of streams composed of multidimensional points. Few studies have focused on clustering streaming time series, which are characterized by being collections of observations sampled sequentially along time...

[72] [Exact indexing of dynamic time warping](#)

Eamonn J. Keogh and C. Ratanamahatana. Knowledge and Information Systems, 2002. 2730 citations.

0% Topic Match

No summary or abstract available

[73] [A review on time series data mining](#)

Tak-Chung Fu. Eng. Appl. Artif. Intell., 2011. 1503 citations.

0% Topic Match

No summary or abstract available

[74] [Accelerating Similarity Search for Elastic Measures: A Study and New Generalization of Lower Bounding Distances](#)

John Paparrizos, ..., and M. Franklin. Proc. VLDB Endow., 2023. 21 citations.

0% Topic Match

Abstract: Similarity search is a core analytical task, and its performance critically depends on the choice of distance measure. For time-series querying, elastic measures achieve state-of-the-art accuracy but are computationally expensive. Thus, fast lower bounding (LB) measures prune unnecessary comparisons with elastic distances to accelerate similarity search. Despite decades of attention, there has never been a study to assess the progress in this area. In addition, the research has disproportionately focused on one popular elastic measure, while other accurate measures have received little or no attention. Therefore, there is merit in developing a framework to accumulate knowledge from previously developed...

[75] [An Interpretable Time Series Clustering Neural Network Based on Shape Feature Extraction](#)

Weide Li, ..., and Zhihe Zhang. Int. J. Pattern Recognit. Artif. Intell., 2022. 1 citations.

0% Topic Match

Abstract: Time series is a very common but important data type. A large number of time series data are generated in various professional research fields and daily life. Although there are many models being developed to deal with time series, the cluster methods for time series are insufficient and need to improve. This paper is focused on time series clustering, which uses deep learning approach to discover the shape characteristics of time series. We establish a new neural network model of time series clustering to jointly optimize the representation learning and clustering tasks of time series. Focusing on shape features with...

[76] [A robust alternating least squares K-means clustering approach for times series using dynamic time warping dissimilarities.](#)

J. F. Vera-Vera and J. A. Roldán-Nofuentes. Mathematical biosciences and engineering : MBE, 2024. 8 citations.

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Abstract: Time series clustering is a usual task in many different areas. Algorithms such as K-means and model-based clustering procedures are used relating to multivariate assumptions on the datasets, as the consideration of Euclidean distances, or a probabilistic distribution of the observed variables. However, in many cases the observed time series are of unequal length and/or there is missing data or, simply, the time periods observed for the series are not comparable between them, which does not allow the direct application of these methods. In this framework, dynamic time warping is an advisable and well-known elastic dissimilarity procedure, in particular when...

[77] [DTW+S: Shape-based Comparison of Time-series with Ordered Local Trend](#)

A. Srivastava. ArXiv, 2023. 2 citations.

0% Topic Match

Abstract: Measuring distance or similarity between time-series data is a fundamental aspect of many applications including classification, clustering, and ensembling/alignment. Existing measures may fail to capture similarities among local trends (shapes) and may even produce misleading results. Our goal is to develop a measure that looks for similar trends occurring around similar times and is easily interpretable for researchers in applied domains. This is particularly useful for applications where time-series have a sequence of meaningful local trends that are ordered, such as in epidemics (a surge to an increase to a peak to a decrease). We propose a novel measure, DTW+S,...

[78] [Fast Global Alignment Kernels](#)

Marco Cuturi. Unknown journal, 2011. 393 citations.

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No summary or abstract available

[79] [Pigeonring: A Principle for Faster Thresholded Similarity Search](#)

Jianbin Qin and Chuan Xiao. ArXiv, 2018. 22 citations.

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Abstract: The pigeonhole principle states that if n items are contained in m boxes, then at least one box has no more than n/m items. It is utilized to solve many data management problems, especially for thresholded similarity searches. Despite many pigeonhole principle-based solutions proposed in the last few decades, the condition stated by the principle is weak. It only constrains the number of items in a single box. By organizing the boxes in a ring, we propose a new principle, called the pigeonring principle, which constrains the number of items in multiple boxes and yields stronger conditions. To utilize...

[80] [Modified Dynamic Time Warping for Hierarchical Clustering](#)

M. Sammour, ..., and Rosmayati Mohamed. International Journal on Advanced Science, Engineering and Information Technology, 2019. 5 citations.

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Abstract: Time series clustering is the process of grouping sequential correspondences in similar clusters. The key feature behind clustering time series data lies on the similarity/distance function used to identify the sequential matches. Dynamic Time Warping (DTW) is one of the common distance measures that have demonstrated competitive results compared to other functions. DTW aims to find the shortest path in the process of identifying sequential matches. DTW relies on dynamic programming to obtain the shortest path where the smaller distance is being computed. However, in the case of equivalent distances, DTW is selecting the path randomly. Hence, the selection could...

[81] [Investigating impacts of COVID-19 on urban mobility and emissions](#)

Duo Li and Joan Lasenby. Cities (London, England), 2023. 18 citations.

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Abstract: The COVID-19 pandemic has severely impacted human activities in a way never documented in modern history. The prevention policies and measures have abruptly changed well-established urban mobility patterns. In this context, we exploit different sources of urban mobility data to gain insights into the effects of restrictive policies on the daily mobility and exhaust emissions in pandemic and post-pandemic periods. Manhattan, the most densely populated borough in New York City, is chosen as the study area. We collect data generated by taxis, sharing bikes, and road detectors between 2019 and 2021, and estimate exhaust emissions using the COPERT (Computer Programme...

[82] [Quantitative Evaluation of Passenger Flow at Metro Stations Based on Coarse-to-Fine Analysis](#)

Yunjie Zhang, ..., and Wei Tu. 2023 30th International Conference on Geoinformatics, 2023. 0 citations.

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Abstract: The dense urban metro network plays an important role in urban transportation, and it is becoming increasingly important to improve the operation and management of metro facilities. The monitoring and management of passenger flow is a main concern in metro operation, and the reliable analysis of passenger flow can greatly improve the operational efficiency and safety of a metro station. Therefore, using the long-time in-and-out smart card data of Shenzhen metro stations, this paper proposes a Coarse-to-Fine passenger flow analysis method for the characterization of passenger flow on multiple time scales. This method, which is proposed from a new perspective...

[83] [Feature Construction Using Persistence Landscapes for Clustering Noisy IoT Time Series](#)

Renjie Chen and N. Ravishanker. Future Internet, 2023. 1 citations.

0% Topic Match

Abstract: With the advancement of IoT technologies, there is a large amount of data available from wireless sensor networks (WSN), particularly for studying climate change. Clustering long and noisy time series has become an important research area for analyzing this data. This paper proposes a feature-based clustering approach using topological data analysis, which is a set of methods for finding topological structure in data. Persistence diagrams and landscapes are popular topological summaries that can be used to cluster time series. This paper presents a framework for selecting an optimal number of persistence landscapes, and using them as features in an unsupervised...

[84] [Barycentre Averaging for the Move-Split-Merge Time Series Distance Measure](#)

Christopher Holder, ..., and Anthony J. Bagnall. Unknown journal, 2023. 3 citations.

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Abstract: Distance functions play a core role in many time series machine learning algorithms for tasks such as clustering, classification and regression. Time series often require bespoke distance functions because small offsets in time can lead to large distances between series that are conceptually similar. Elastic distances compensate for misalignment by creating a path through a cost matrix by warping and/or editing time series. Time series are most commonly clustered with partitional algorithms such as k-means and k-medoids using elastic distance measures such as Dynamic Time Warping (DTW). The distance is used to assign cases to the closest...

[85] [Data Series Similarity Using Correlation-Aware Measures](#)

Katsiaryna Milylenka, ..., and Themis Palpanas. Proceedings of the 29th International Conference on Scientific and Statistical Database Management, 2017. 17 citations.

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No summary or abstract available

[86] [TISIS : Trajectory Indexing for Smilarity Search](#)

Sara Jarrad, ..., and S. Gançarski. ArXiv, 2024. 0 citations.

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Abstract: Social media platforms enable users to share diverse types of information, including geolocation data that captures their movement patterns. Such geolocation data can be leveraged to reconstruct the trajectory of a user's visited Points of Interest (POIs). A key requirement in numerous applications is the ability to measure the similarity between such trajectories, as this facilitates the retrieval of trajectories that are similar to a given reference trajectory. This is the main focus of our work. Existing methods predominantly rely on applying a similarity function to each candidate trajectory to identify those that are sufficiently similar. However, this approach becomes...

[87] [Using Dynamic Time Warping to Find Patterns in Time Series](#)

D. Berndt and J. Clifford. Unknown journal, 1994. 4274 citations.

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No summary or abstract available

[88] [Time-Series K-means in Causal Inference and Mechanism Clustering for Financial Data](#)

S. Bo and Minheng Xiao. Proceedings of the 2024 7th International Conference on Computer Information Science and Artificial Intelligence, 2022. 3 citations.

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Abstract: This paper examines the application of the time series k-means (TS-k-means) algorithm in the analysis of financial time series data. It employs dynamic time warping (DTW) as the metric and addresses the limitation of traditional clustering methods in capturing time dynamics. When combined with the additive noise Model Mixed Model (ANM-MM) architecture, this strategy not only enhances the accuracy of causal inference, but also enables clustering based on intrinsic generation mechanisms, thereby providing solid support for an in-depth understanding of complex financial time series data. The combination of TS-k-means and ANM-MM has been demonstrated to exhibit excellent performance in capturing...

[89] [Categorization of crowd-sensing streaming data for contextual characteristic detection](#)

Philipp Kisters, ..., and Janick Edinger. Journal of Smart Cities and Society, 2023. 1 citations.

0% Topic Match

Abstract: The growing reliance on large wireless sensor networks, potentially consisting of hundreds of nodes, to monitor real-world phenomena inevitably results in large, complex datasets that become increasingly difficult to process using traditional methods. The inadvertent inclusion of anomalies in the dataset, resulting from the inherent characteristics of these networks, makes it difficult to isolate interesting events from erroneous measurements. Simultaneously, improvements in data science methods, as well as increased accessibility to powerful computers, lead to these techniques becoming more applicable to everyday data mining problems. In addition to being able to process large amounts of complex streaming data, a wide...

[90] [Indexing spatio-temporal trajectories with Chebyshev polynomials](#)

Yuhan Cai and R. Ng. Unknown journal, 2004. 360 citations.

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No summary or abstract available

[91] [Time Series Overlapping Clustering Based on Link Community Detection](#)

Yasamin Ghahremani and Babak Amiri. IEEE Access, 2024. 0 citations.

0% Topic Match

Abstract: Given the nature of time series and their vast applications, it is essential to find clustering algorithms that depict their real-life properties. Among the features that can hugely effect the options available for time series are overlapping and hierarchical properties. In this paper a novel approach to analyze time series with such features is introduced. Using the two concepts of network construction and link community detection, we have attempted to analyze and identify the mentioned properties of time series using data that is often gathered first hand. The proposed algorithm has been applied using both recent and common similarity measures...

[92] [Ieee Transactions on Knowledge and Data Engineering, Manuscript Id 1 a General Framework for Density Based Time Series Clustering Exploiting a Novel Admissible Pruning Strategy](#)

Nurjahan Begum, ..., and Eamonn J. Keogh. Journal Not Provided, Unknown year. 7 citations.

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No summary or abstract available

[93] [Robust and fast similarity search for moving object trajectories](#)

Lei Chen, ..., and Vincent Oria. Unknown journal, 2005. 1407 citations.

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No summary or abstract available

[94] [Distributed Trajectory Similarity Search](#)

Dong Xie, ..., and J. M. Phillips. Proc. VLDB Endow., 2017. 141 citations.

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No summary or abstract available

[95] [Fast Similarity Search in the Presence of Noise, Scaling, and Translation in Time-Series Databases](#)

R. Agrawal, ..., and Kyuseok Shim. Unknown journal, 1995. 797 citations.

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No summary or abstract available

[96] [Customer Behavior Clustering Based on Balance History Using Dynamic Time Warping Distance](#)

P. D. Hung and Do Quang Dat. International Journal of Machine Learning and Computing, 2020. 1 citations.

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Abstract: Abstract —Customer clustering, the division of customers into different groups, is a classical problem. It is especially important in banking as it serves multiple purposes in marketing, risk management, etc. Therefore, it has attracted the use of many modern machine learning models and techniques. But currently, most of them are only making use of "static" customer information. This paper proposes a new approach for customer clustering in banking based on the customers' balance history. Basic Dynamic Time Warping (DTW) distance and Soft-DTW (SDTW) distance, an advanced form, are used to measure the difference between customers. To which, the two...

[97] [Dataset Discovery via Line Charts](#)

Daomin Ji, ..., and J. Culpepper. 2025 IEEE 41st International Conference on Data Engineering (ICDE), 2024. 0 citations.

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Abstract: Line charts are a valuable tool for data analysis and exploration, distilling essential insights from a dataset. However, access to the underlying data used to create a line chart is rarely readily available. In this paper, we explore a novel dataset discovery problem, dataset discovery via line charts, focusing on the use of line charts as queries to discover datasets within a large data repository that are capable of generating similar line charts. To solve this problem, we propose a novel approach called Fine-grained Cross-modal Relevance Learning Model (FCM), which aims to estimate the relevance between a line chart and...

[98] [Time series clustering using stochastic and deterministic influences](#)

Ricardo Rios, ..., and Mirlei Silva. Int. J. Comput. Sci. Eng., 2020. 2 citations.

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Abstract: Time series clustering aims at designing methods to extract patterns from temporal data in order to organise series according to their similarities. In general, most of researches either perform a preprocessing step to convert time series into attribute-value matrices to be later analysed by traditional clustering methods, or apply measures specifically designed to compute the similarity among time series. We noticed two main issues in such studies: 1) clustering methods do not take into account stochastic and deterministic influences inherent in real-world time series; 2) similarity measures tend to look for recurrent patterns, which may not be available in stochastic...

- [99] [Discovering similar multidimensional trajectories](#)
M. Vlachos, ..., and G. Kollios. Proceedings 18th International Conference on Data Engineering, 2002. 1654 citations.
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No summary or abstract available
- [100] [Efficient Retrieval of Similar Time Sequences Using DFT](#)
Davood Rafiei and A. Mendelzon. ArXiv, 1998. 663 citations.
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Abstract: We propose an improvement of the known DFT-based indexing technique for fast retrieval of similar time sequences. We use the last few Fourier coefficients in the distance computation without storing them in the index since every coefficient at the end is the complex conjugate of a coefficient at the beginning and as strong as its counterpart. We show analytically that this observation can accelerate the search time of the index by more than a factor of two. This result was confirmed by our experiments, which were carried out on real stock prices and synthetic data.
- [101] [BrainEx: Interactive Visual Exploration and Discovery of Sequence Similarity in Brain Signals](#)
Alicia Howell-Munson, ..., and Rodica Neamtu. Proceedings of the ACM on Human-Computer Interaction, 2022. 1 citations.
0% Topic Match
Abstract: Technology advances and lower equipment costs are enabling non-invasive, convenient recording of brain data outside of clinical settings in more real-world environments, and by non-experts. Despite the growing interest in and availability of brain signal datasets, most analytical tools are made for experts in the specific device technology, and have rigid constraints on the type of analysis available. We developed BrainEx to support interactive exploration and discovery within brain signals datasets. BrainEx takes advantage of algorithms that enable fast exploration of complex, large collections of time series data, while being easy to use and learn. This system enables researchers to...
- [102] [A Survey of Temporal Knowledge Discovery Paradigms and Methods](#)
J. Roddick and M. Spiliopoulou. IEEE Trans. Knowl. Data Eng., 2002. 481 citations.
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No summary or abstract available
- [103] [ShapeDBA: Generating Effective Time Series Prototypes using ShapeDTW Barycenter Averaging](#)
Ali Ismail-Fawaz, ..., and G. Forestier. ArXiv, 2023. 7 citations.
0% Topic Match
Abstract: Time series data can be found in almost every domain, ranging from the medical field to manufacturing and wireless communication. Generating realistic and useful exemplars and prototypes is a fundamental data analysis task. In this paper, we investigate a novel approach to generating realistic and useful exemplars and prototypes for time series data. Our approach uses a new form of time series average, the ShapeDTW Barycentric Average. We therefore turn our attention to accurately generating time series prototypes with a novel approach. The existing time series prototyping approaches rely on the Dynamic Time Warping (DTW) similarity measure such as DTW...
- [104] [Fast and synchronized multidimensional similarity measure for trajectories: integrating space, time, and semantic information](#)
Juqing Liu, ..., and Chengye Zhang. International Journal of Digital Earth, 2025. 0 citations.
0% Topic Match
Abstract: ABSTRACT While the rich semantic information in trajectories enhances data mining potential, it simultaneously complicates similarity measures. Existing multidimensional similarity methods face two challenges: (1) high computational complexity of $O(n \times m \times k)$, which limits large-scale applicability, and (2) isolated handling of spatial-temporal-semantic dimensions with inadequate semantic hierarchy modeling. This paper proposes FasMultiSIM, a fast and synchronized multidimensional similarity measure that integrates spatial, temporal, and semantic dimensions. First, a spatiotemporal grid model (rSTGM) is constructed by extending rHEALPix DGGS with temporal dimensions, enabling FasMultiSIM to achieve fast multidimensional similarity computations with time complexity of $O((n + m) \times k)$. In addition,...
- [105] [Dimensionality Reduction for Fast Similarity Search in Large Time Series Databases](#)
Eamonn J. Keogh, ..., and S. Mehrotra. Knowledge and Information Systems, 2001. 1658 citations.
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No summary or abstract available
- [106] [Fast Time Sequence Indexing for Arbitrary Lp Norms](#)
Byoung-Kee Yi and C. Faloutsos. Unknown journal, 2000. 774 citations.
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No summary or abstract available
- [107] [A Principled Distributional Approach to Trajectory Similarity Measurement and its Application to Anomaly Detection](#)
Yufan Wang, ..., and Yuanyi Shang. J. Artif. Intell. Res., 2024. 6 citations.
0% Topic Match
Abstract: This paper aims to solve two enduring challenges in existing trajectory similarity measures: computational inefficiency and the absence of the 'uniqueness' property that should be guaranteed in a distance function: $\text{dist}(X, Y) = 0$ if and only if $X = Y$, where X and Y are two trajectories. In this work, we present a novel approach utilizing a distributional kernel for trajectory representation and similarity measurement, based on the kernel mean embedding framework. It is the very first time a distributional kernel is used for trajectory representation and similarity measurement. Our method does not rely on point-to-point distances...
- [108] [Exact and Efficient Similar Subtrajectory Search: Integrating Constraints and Simplification](#)
Liwei Deng, ..., and Kai Zheng. 2025 IEEE 41st International Conference on Data Engineering (ICDE), 2025. 2 citations.
0% Topic Match
Abstract: Similar subtrajectory search (SimSub) aims to find a subtrajectory (i.e., a segment) from a data trajectory (the trajectory to be queried) that closely resembles the query trajectory. Compared with similar trajectory search, SimSub can capture finer-grained similarity and is vital for various trajectory analysis tasks, such as trajectory clustering and join. However, SimSub may return a subtrajectory with extremely limited length, e.g., a single point, which may not align with the expectations of real-world applications. To solve this issue, we propose a constrained SimSub (cSimSub) problem, where the length of the returned subtrajectory must be greater than or equal to...
- [109] [Quantifying the Spatiotemporal Dynamics of Engineered Cardiac Microbundles](#)
Hiba Kobeissi, ..., and Emma Lejeune. ArXiv, 2026. 0 citations.
0% Topic Match
Abstract: Brightfield time-lapse imaging is widely used in cardiac tissue engineering, yet the absence of standardized, interpretable analytical frameworks limits reproducibility and cross-platform comparison. We present an open, scalable computational pipeline for quantifying spatiotemporal contractile dynamics in microscopy videos of human induced pluripotent stem cell-derived cardiac microbundles. Building on our open-source tools "MicroBundleCompute" and "MicroBundlePillarTrack," we define a suite of 16 interpretable structural, functional, and spatiotemporal metrics that capture tissue deformation, synchrony, and heterogeneity. The framework integrates full-field displacement tracking, strain reconstruction, spatial registration, dimensionality reduction, and topology-based vector-field analysis within a unified workflow. Applied to a dataset of 670 cardiac microbundles spanning 20 experimental...
- [110] [Dynamic spatio-temporal pattern discovery: a novel grid and density-based clustering algorithm](#)
S. Meshram and Kishor P Wagh. IAES International Journal of Artificial Intelligence (IJ-AI), 2025. 0 citations.
0% Topic Match
Abstract: Clustering is a robust machine-learning technique for exploration of patterns based on similarity of elements over multidimensional data. Spatio-temporal clustering aims to identify target objects to mine spatial and temporal dimensions for patterns, regularity, and trends. It has been applied in human-centric applications, such as recommendation systems, urban development and planning, clustering of criminal activities, traffic planning, and epidemiology to identify the extent of

disease spread. Although the existing research work in the field of clustering relies widely on partition and densitybased methods, no major work has been carried out to handle the spatiotemporal dimension and understand the dynamics...

[111] [Fast subsequence matching in time-series databases](#)

Christos Faloutsos, ..., and Y. Manolopoulos. Unknown journal, 1994. 2078 citations.

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No summary or abstract available

[112] [Clustering of Temporal and Visual Data: Recent Advancements](#)

Priyanka Mudgal. Data, 2026. 1 citations.

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Abstract: Clustering plays a central role in uncovering latent structure within both temporal and visual data. It enables critical insights in various domains including healthcare, finance, surveillance, autonomous systems, and many more. With the growing volume and complexity of time-series and image-based datasets, there is an increasing demand for robust, flexible, and scalable clustering algorithms. Although these modalities differ—time-series being inherently sequential and vision data being spatial—they exhibit common challenges such as high dimensionality, noise, variability in alignment and scale, and the need for interpretable groupings. This survey presents a comprehensive review of recent advancements in clustering methods that are adaptable...

[113] [Index-based Most Similar Trajectory Search](#)

E. Frentzos, ..., and Y. Theodoridis. 2007 IEEE 23rd International Conference on Data Engineering, 2007. 229 citations.

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No summary or abstract available

[114] [Modeling Bank Systemic Risk of Emerging Markets under Geopolitical Shocks: Empirical Evidence from BRICS Countries](#)

Haibo Wang. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: In this study, we introduce an analytics framework, the Bank Risk Interlinkage with Dynamic Graph and Event Simulations (BRIDGES), to capture the systemic risks associated with the growing economic influence of the BRICS nations. This framework includes a Dynamic Time Warping (DTW) method to construct a dynamic network of 551 BRICS banks with their annual balance sheet data from 2008 to 2024; a trend analysis in risk ratios to detect shifts in banks' behavior; a Temporal Graph Neural Network (TGNN) to detect anomalous changes in the bank network's structural relationships; and Agent-Based Model (ABM) simulations to measure the impact of anomalous...

[115] [Quantum Time Series Similarity Measures and Quantum Temporal Kernels](#)

Vanio Markov, ..., and D. Fry. ArXiv, 2023. 2 citations.

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Abstract: This article presents a quantum computing approach to designing of similarity measures and kernels for classification of stochastic symbolic time series. In the area of machine learning, kernels are important components of various similarity-based classification, clustering, and regression algorithms. An effective strategy for devising problem-specific kernels is leveraging existing generative models of the example space. In this study we assume that a quantum generative model, known as quantum hidden Markov model (QHMM), describes the underlying distributions of the examples. The sequence structure and probability are determined by transitions within model's density operator space. Consequently, the QHMM defines a mapping from...

[116] [Financial time series augmentation using transformer based GAN architecture](#)

Andrzej Podobiński and J. Chudziak. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Time-series forecasting is a critical task across many domains, from engineering to economics, where accurate predictions drive strategic decisions. However, applying advanced deep learning models in challenging, volatile domains like finance is difficult due to the inherent limitation and dynamic nature of financial time series data. This scarcity often results in sub-optimal model training and poor generalization. The fundamental challenge lies in determining how to reliably augment scarce financial time series data to enhance the predictive accuracy of deep learning forecasting models. Our main contribution is a demonstration of how Generative Adversarial Networks (GANs) can effectively serve as a data...

[117] [Feature Enhanced Spatial-Temporal Trajectory Similarity Computation](#)

Silin Zhou, ..., and Lisi Chen. Data Science and Engineering, 2024. 3 citations.

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[118] [Efficient Similarity Search in Sequence Databases](#)

Agrawal Christos Faloutsos and Arun SwamiIBM. Journal Not Provided, 1993. 141 citations.

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No summary or abstract available

[119] [Tree-based methods for clustering time series using domain-relevant attributes](#)

Mahsa Ashouri, ..., and Chor-yiu Sin. Journal of Business Analytics, 2019. 14 citations.

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Abstract: ABSTRACT We propose two methods for time-series clustering that capture temporal information (trend, seasonality, autocorrelation) and domain-relevant cross-sectional attributes. The methods are based on model-based partitioning (MOB) trees and can be used as automated yet transparent tools for clustering large collections of time series. We address the challenge of using common time-series models in MOB by instead utilising least squares regression. We propose two methods. The single-step method clusters series using trend, seasonality, lags and domain-relevant cross-sectional attributes. The two-step method first clusters by trend, seasonality and cross-sectional attributes, and then clusters the residuals by autocorrelation and domain-relevant attributes. Both...

[120] [Similarity Search on Time Series Based on Threshold Queries](#)

Johannes Aßfalg, ..., and M. Renz. Unknown journal, 2006. 105 citations.

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No summary or abstract available

[121] [Towards plausibility in time series counterfactual explanations](#)

Marcin Kostrzewa, ..., and Maciej Zikeba. ArXiv, 2026. 0 citations.

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Abstract: We present a new method for generating plausible counterfactual explanations for time series classification problems. The approach performs gradient-based optimization directly in the input space. To enforce plausibility, we integrate soft-DTW (dynamic time warping) alignment with k -nearest neighbors from the target class, which effectively encourages the generated counterfactuals to adopt a realistic temporal structure. The overall optimization objective is a multi-faceted loss function that balances key counterfactual properties. It incorporates losses for validity, sparsity, and proximity, alongside the novel soft-DTW-based plausibility component. We conduct an evaluation of our method against several strong reference approaches, measuring the key properties of the...

[122] [A Comprehensive Comparison of Distance Measures for Time Series Classification](#)

Tomasz Górecki and Paweł Piasecki. Springer Proceedings in Mathematics & Statistics, 2019. 10 citations.

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No summary or abstract available

[123] [Indexable PLA for Efficient Similarity Search](#)

Qiuxia Chen, ..., and J. Yu. Unknown journal, 2007. 147 citations.

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[124] [Similarity Learning for Time Series Classification](#)

Maria-Irina Nicolae, ..., and M. Sebban. ArXiv, 2016. 4 citations.

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Abstract: Multivariate time series naturally exist in many fields, like energy, bioinformatics, signal processing, and finance. Most of these applications need to be able to compare these structured data. In this context, dynamic time warping (DTW) is probably the most common comparison measure. However, not much research effort has been put into improving it by learning. In this paper, we propose a novel method for learning similarities based on DTW, in order to improve time series classification. Making use of the uniform stability framework, we provide the first theoretical guarantees in the form of a generalization bound for linear classification. The...

[125] [Time-Series Similarity and Clustering of Producer Share Dynamics in Agrifood Markets: Evidence from Origin–Destination Price Relationships](#)

Elena Sánchez-Arnau, ..., and E. A. Sánchez-Pérez. Mathematics, 2026. 0 citations.

0% Topic Match

Abstract: Producer share indicators summarize how value is distributed along agrifood supply chains, yet their temporal dynamics remain difficult to compare across products and periods. This paper proposes a reproducible time-series analytics framework to characterize and group producer-share trajectories derived from paired origin–destination price series. We compute producer share time series for a set of agrifood products and quantify similarity using complementary measures capturing co-movement and shape, including Pearson-correlation-based proximity and Euclidean distance on standardized representations. To reduce dimensionality and mitigate noise, we apply principal component analysis and perform unsupervised clustering (k-means) to identify classes of products exhibiting comparable producer-share dynamics....

[126] [SpADe: On Shape-based Pattern Detection in Streaming Time Series](#)

Yueguo Chen, ..., and A. Tung. 2007 IEEE 23rd International Conference on Data Engineering, 2007. 191 citations.

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No summary or abstract available

[127] [Having It Both Ways: Single Trajectory Embedding for Similarity Computation with Pairwise Learning](#)

Jianing Si, ..., and Shangguang Wang. 2025 IEEE 41st International Conference on Data Engineering (ICDE), 2025. 0 citations.

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Abstract: Trajectory similarity measure is a fundamental component in trajectory databases, supporting many down-stream trajectory tasks. Existing similarity functions often exhibit unacceptable time complexities, hampering their efficiency for real-world scenarios. To address this limitation, learning-based approximation techniques utilizing trajectory embeddings have been proposed. However, creating a robust embedding model presents challenges, including the lack of direct involvement in the computational similarity process, adherence to non-metric similarity spaces, and the integration of precise similarity computation alignments. To address these challenges, we introduce DTisT, a novel embedding framework that enhances trajectory embeddings by pairwise learning from dual-trajectory input models. DTisT not only captures...

[128] [Learned Data-aware Image Representations of Line Charts for Similarity Search](#)

Yuyu Luo, ..., and Leixian Shen. Proceedings of the ACM on Management of Data, 2023. 28 citations.

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No summary or abstract available

[129] [Time-series similarity problems and well-separated geometric sets](#)

B. Bollobás, ..., and H. Mannila. Unknown journal, 1997. 130 citations.

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[130] [Modeling and predicting individual transitions within the homelessness system](#)

Khandker Sadia Rahman and C. Chelmiss. Social Network Analysis and Mining, 2023. 0 citations.

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[131] [From Interaction to Demonstration Quality in Virtual Reality: Effects of Interaction Modality and Visual Representation on Everyday Tasks](#)

Robin Beierling, ..., and Anna-Lisa Vollmer. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Virtual Reality (VR) is increasingly used for training and demonstration purposes including a variety of applications ranging from robot learning to rehabilitation. However, the choice of input device and its visualization might influence workload and thus user performance leading to suboptimal demonstrations or reduced training effects. This study investigates how different VR input configurations - motion capture gloves, controllers with hand visualization, and controllers with controller visualization - affect user experience and task execution, with the goal of identifying which configuration is best suited for which type of task. Participants performed various kitchen-related activities of daily living (ADLs), including object...

[132] [Using Signature Files for Querying Time-Series Data](#)

Henrik André-Jönsson and D. Badal. Unknown journal, 1997. 105 citations.

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No summary or abstract available

[133] [Clustering Individuals Based on Similarity in Idiographic Factor Loading Patterns](#)

Cara J Arizmendi and Kathleen M Gates. Multivariate Behavioral Research, 2024. 1 citations.

0% Topic Match

Abstract: Abstract Idiographic measurement models such as p-technique and dynamic factor analysis (DFA) assess latent constructs at the individual level. These person-specific methods may provide more accurate models than models obtained from aggregated data when individuals are heterogeneous in their processes. Developing clustering methods for the grouping of individuals with similar measurement models would enable researchers to identify if measurement model subtypes exist across individuals as well as assess if the different models correspond to the same latent concept or not. In this paper, methods for clustering individuals based on similarity in measurement model loadings obtained from time series data are...

[134] [Dynamic programming algorithm optimization for spoken word recognition](#)

Hiroaki Sakoe. IEEE Transactions on Acoustics, Speech, and Signal Processing, 1978. 6843 citations.

0% Topic Match

No summary or abstract available

[135] [Machine and deep learning as tools for monitoring forest recovery from disturbance in the western United States](#)

J. Wickham, ..., and Jeremy Baynes. International Journal of Remote Sensing, 2026. 0 citations.

0% Topic Match

Abstract: ABSTRACT Changing disturbance regimes in the western United States have raised concerns that forests may not recover following disturbance. We used 20 years of land cover change data and 23 years of summer maximum NDVI to assess forest recovery. The land cover change data were used to identify where forest was lost early on in the 20-year period and the NDVI data were used to monitor recovery at disturbed locations. Two different target classifications were developed to categorize post-disturbance NDVI trends and convolutional neural networks were used to detect outliers – locations for which the NDVI response did not fit...

[136] [StretchTime: Adaptive Time Series Forecasting via Symplectic Attention](#)

Yubin Kim, ..., and Jiecheng Lu. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Transformer architectures have established strong baselines in time series forecasting, yet they typically rely on positional encodings that assume uniform, index-based temporal progression. However, real-world systems, from shifting financial cycles to elastic biological rhythms, frequently exhibit "time-warped" dynamics where the effective flow of time decouples from the sampling index. In this work, we first formalize this misalignment and prove that rotary position embedding (RoPE) is mathematically incapable of representing non-affine temporal warping. To address this, we propose Symplectic Positional Embeddings (SyPE), a learnable encoding framework derived from Hamiltonian mechanics. SyPE strictly generalizes RoPE by extending the rotation group $\mathrm{SO}(2)$ to the symplectic...

[137] [Toward Secure Trajectory Similarity Range Query Under Multiuser Setting](#)

Ningning Cui, ..., and Hong Zhong. IEEE Internet of Things Journal, 2025. 0 citations.

0% Topic Match

Abstract: The widespread availability of similarity queries over trajectory data has led to numerous real-world applications, such as traffic management and path planning. With the proliferation of trajectory data, data owners often outsource storage and computation tasks to the cloud due to limited computing and storage resources. However, this scenario raises sharp security concerns, where it is critical to ensure both the integrity of query results and privacy during query processing. Furthermore, most existing works assume a single-user setting where all query users share the same key, which may lead to query privacy leakage. Therefore, in this article, we take the...

[138] [Log2Motion: Biomechanical Motion Synthesis from Touch Logs](#)

MichaB Patryk Miazga, ..., and Patrick EbeProceedings of the 2026 CHI Conference on Human Factors in Computing Systems, 2026. 0 citations.

0% Topic Match

Abstract: Touch data from mobile devices are collected at scale but reveal little about the interactions that produce them. While biomechanical simulations can illuminate motor control processes, they have not yet been developed for touch interactions. To close this gap, we propose a novel computational problem: synthesizing plausible motion directly from logs. Our key insight is a reinforcement learning-driven musculoskeletal forward simulation that generates biomechanically plausible motion sequences consistent with events recorded in touch logs. We achieve this by integrating a software emulator into a physics simulator, allowing biomechanical models to manipulate real applications in real-time. Log2Motion produces rich syntheses of...

[139] [Efficient Similarity Search In Sequence Databases](#)

R. Agrawal, ..., and A. Swami. Unknown journal, 1993. 2205 citations.

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No summary or abstract available

[140] [Explainable AI for Binary Black Hole Light Curve Classification via Feature-Weighted Embeddings](#)

Vicente Echegaray and J. Cuadra. 2025 15th IEEE International Conference on Pattern Recognition Systems (ICPRS), 2025. 0 citations.

0% Topic Match

Abstract: Classifying binary black-hole (BBH) systems from noisy, irregular light curves poses a significant challenge in astrophysics. This work introduces an explainable AI pipeline designed to overcome this roadblock by comparing accretion rate time series from a target source with a catalog of simulations. Our methodological approach involves three key stages: (i) automated feature extraction of accretion-rate time series using pycatch22; (ii) supervised feature-weight learning via multi-output Random Forests to generate interpretable importance scores; and (iii) nearest-neighbor retrieval in a weighted-embedding space for inferring the mass ratio (q) and eccentricity (e) of the BBHs. This pipeline effectively leverages the scalability of...

[141] [Time-series clustering - A decade review](#)

S. Aghabozorgi, ..., and Teh Ying Wah. Inf. Syst., 2015. 1708 citations.

Not measured Topic Match

No summary or abstract available

[142] [A New Time Series Similarity Measure and Its Smart Grid Applications](#)

R. Yuan, ..., and Julian Lemos-Vinasco. ArXiv, 2023. 2 citations.

Not measured Topic Match

Abstract: Many smart grid applications involve data mining, clustering, classification, identification, and anomaly detection, among others. These applications primarily depend on the measurement of similarity, which is the distance between different time series or subsequences of a time series. The commonly used time series distance measures, namely Euclidean Distance (ED) and Dynamic Time Warping (DTW), do not quantify the flexible nature of electricity usage data in terms of temporal dynamics. As a result, there is a need for a new distance measure that can quantify both the amplitude and temporal changes of electricity time series for smart grid applications, e.g., demand...

[143] [On the Use of Relative Validity Indices for Comparing Clustering Approaches](#)

Luke W. Yerbury, ..., and Lachlan O'Neil. ACM Transactions on Knowledge Discovery from Data, 2024. 5 citations.

Not measured Topic Match

Abstract: Relative Validity Indices (RVIs) such as the Silhouette Width Criterion, Calinski-Harabasz and Davies-Bouldin indices are the most widely used tools for evaluating and optimising clustering outcomes. Traditionally, their ability to rank collections of candidate dataset partitions has been used to guide the selection of the number of clusters and to compare partitions from different clustering algorithms. However, there is a growing trend in the literature to use RVIs when selecting a Similarity Paradigm (SP) for clustering—the combination of normalisation procedure, representation method and distance measure which affects the computation of object dissimilarities used in clustering. Despite the growing prevalence of...

[144] [On the Need for Time Series Data Mining Benchmarks: A Survey and Empirical Demonstration](#)

Eamonn J. Keogh and Shruti Kasetty. Data Mining and Knowledge Discovery, 2002. 1442 citations.

Not measured Topic Match

No summary or abstract available

[145] [A User-Friendly Guide to Using Distance Measures to Compare Time Series in Ecology](#)

Shawn Dove, ..., and D. Murrell. bioRxiv, 2022. 5 citations.

Not measured Topic Match

Abstract: Time series are a critical component of ecological analysis, used to track changes in biotic and abiotic variables. Information can be extracted from the properties of time series for tasks such as classification, clustering, prediction, and anomaly detection. These common tasks in ecological research rely on the notion of (dis-) similarity which can be determined by using distance measures. A plethora of distance measures have been described in the scientific literature, but many of them have not been introduced to ecologists. Furthermore, little is known about how to select appropriate distance measures and the properties they focus on for time-series...

[146] [Benchmarking distance-based partitioning methods for mixed-type data](#)

Efthymios Costa, ..., and Angelos Markos. Advances in Data Analysis and Classification, 2022. 21 citations.

Not measured Topic Match

Abstract: Clustering mixed-type data, that is, observation by variable data that consist of both continuous and categorical variables poses novel challenges. Foremost among these challenges is the choice of the most appropriate clustering method for the data. This paper presents a benchmarking study comparing eight distance-based partitioning methods for mixed-type data in terms of cluster recovery performance. A series of simulations carried out by a full factorial design are presented that examined the effect of a variety of factors on cluster recovery. The amount of cluster overlap, the percentage of categorical variables in the data set, the number of clusters and...

[147] [Hierarchical Clustering for Smart Meter Electricity Loads Based on Quantile Autocovariances](#)

A. Alonso, ..., and C. Ruiz. IEEE Transactions on Smart Grid, 2019. 48 citations.

Not measured Topic Match

Abstract: In order to improve the efficiency and sustainability of electricity systems, most countries worldwide are deploying advanced metering infrastructures, and in particular household smart meters, in the residential sector. This technology is able to record electricity load time series at a very high frequency rates, information that can be exploited to develop new clustering models to group individual households by similar consumptions patterns. To this end, in this work we propose three

hierarchical clustering methodologies that allow capturing different characteristics of the time series. These are based on a set of "dissimilarity" measures computed over different features: quantile auto-covariances, and...

[148] [A white paper on good research practices in benchmarking: The case of cluster analysis](#)

I. Mechelen, ..., and D. Steinley. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 2018. 36 citations.

Not measured Topic Match

Abstract: To achieve scientific progress in terms of building a cumulative body of knowledge, careful attention to benchmarking is of the utmost importance, requiring that proposals of new methods are extensively and carefully compared with their best predecessors, and existing methods subjected to neutral comparison studies. Answers to benchmarking questions should be evidence based, with the relevant evidence being collected through well thought out procedures, in reproducible and replicable ways. In the present paper, we review good research practices in benchmarking from the perspective of the area of cluster analysis. Discussion is given to the theoretical, conceptual underpinnings of benchmarking based on simulated and...

[149] [Analysis of Similarity Measures in Times Series Clustering for the Discovery of Building Energy Patterns](#)

F. Iglesias and W. Kastner. Energies, 2013. 221 citations.

Not measured Topic Match

Abstract: Forecasting and modeling building energy profiles require tools able to discover patterns within large amounts of collected information. Clustering is the main technique used to partition data into groups based on internal and a priori unknown schemes inherent of the data. The adjustment and parameterization of the whole clustering task is complex and submitted to several uncertainties, being the similarity metric one of the first decisions to be made in order to establish how the distance between two independent vectors must be measured. The present paper checks the effect of similarity measures in the application of clustering for discovering representatives...

[150] [TSclust: An R Package for Time Series Clustering](#)

Pablo Montero and J. A. Vilar. Journal of Statistical Software, 2014. 441 citations.

Not measured Topic Match

Abstract: Time series clustering is an active research area with applications in a wide range of fields. One key component in cluster analysis is determining a proper dissimilarity measure between two data objects, and many criteria have been proposed in the literature to assess dissimilarity between two time series. The R package TSclust is aimed to implement a large set of well-established peer-reviewed time series dissimilarity measures, including measures based on raw data, extracted features, underlying parametric models, complexity levels, and forecast behaviors. Computation of these measures allows the user to perform clustering by using conventional clustering algorithms. TSclust also includes...

[151] [The great multivariate time series classification bake off: a review and experimental evaluation of recent algorithmic advances](#)

Alejandro Pasos Ruiz, ..., and A. Bagnall. Data Mining and Knowledge Discovery, 2020. 458 citations.

Not measured Topic Match

Abstract: Time Series Classification (TSC) involves building predictive models for a discrete target variable from ordered, real valued, attributes. Over recent years, a new set of TSC algorithms have been developed which have made significant improvement over the previous state of the art. The main focus has been on univariate TSC, i.e. the problem where each case has a single series and a class label. In reality, it is more common to encounter multivariate TSC (MTSC) problems where the time series for a single case has multiple dimensions. Despite this, much less consideration has been given to MTSC than the univariate...

[152] [Clustering time series from ARMA models with clipped data](#)

A. Bagnall and G. Janacek. Proceedings of the tenth ACM SIGKDD international conference on Knowledge discovery and data mining, 2004. 112 citations.

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[153] [k-Shape: Efficient and Accurate Clustering of Time Series](#)

John Paparrizos and L. Gravano. Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data, 2015. 513 citations.

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No summary or abstract available

[154] [GRAIL: Efficient Time-Series Representation Learning](#)

John Paparrizos and M. Franklin. Proc. VLDB Endow., 2019. 64 citations.

Not measured Topic Match

Abstract: The analysis of time series is becoming increasingly prevalent across scientific disciplines and industrial applications. The effectiveness and the scalability of time-series mining techniques critically depend on design choices for three components responsible for (i) representing; (ii) comparing; and (iii) indexing time series. Unfortunately, these components have to date been investigated and developed independently, often resulting in mutually incompatible methods. The lack of a unified approach has hindered progress towards fast and accurate analytics over massive time-series collections. To address this major drawback, we present GRAIL, a generic framework to learn compact time-series representations that preserve the properties of a...

[155] [A global averaging method for dynamic time warping, with applications to clustering](#)

F. Petitjean, ..., and P. Gançarski. Pattern Recognit., 2011. 1101 citations.

Not measured Topic Match

No summary or abstract available

[156] [Algorithm 1061: tsdistances: A High-Performance Python Library for Time Series Distances with GPU Support](#)

A. Azzari, ..., and Pietro Sala. ACM Transactions on Mathematical Software, 2026. 0 citations.

Not measured Topic Match

Abstract: Time series distance measures are fundamental in numerous domains, including finance, healthcare, and signal processing, enabling crucial tasks such as pattern recognition, anomaly detection, and predictive modeling. However, many applications require computing distances between all pairs of time series in large datasets, a computationally intensive task that can become a significant bottleneck in analysis pipelines. The tsdistances library is a high-performance Python package designed for computing distances between time series, with GPU support for accelerated processing. This article introduces tsdistances and its key features, focusing on the implementation of elastic distance algorithms and their optimizations. We present both CPU and...

[157] [A New Lower Bounding Paradigm and Tighter Lower Bounds for Elastic Similarity Measures](#)

Zemin Chao, ..., and Hongzhi Wang. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Elastic similarity measures are fundamental to time series similarity search because of their ability to handle temporal misalignments. These measures are inherently computationally expensive, therefore necessitating the use of lower bounds to prune unnecessary comparisons. This paper proposes a new \emph{Bipartite Graph Edge-Cover Paradigm} for deriving lower bounds, which applies to a broad class of elastic similarity measures. This paradigm formulates lower bounding as a vertex-weighting problem on a weighted bipartite graph induced from the input time series. Under this paradigm, most of the existing lower bounds of elastic similarity measures can be viewed as simple instantiations. We further propose...

[158] [TSQ: An Optimized Framework for Efficiently Answering Time Series Queries](#)

Feifan Pu and Jianqiu Xu. Data Science and Engineering, 2025. 3 citations.

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[159] [HPS: A novel heuristic hierarchical pruning strategy for dynamic top-k trajectory similarity query](#)

Tianyi Gu, ..., and Lei Zhao. Inf. Process. Manag., 2024. 3 citations.

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- [160] [Spatiotemporal dynamics of human mobility after lifting of COVID-19 control measures in China: evidence from Nanjing City's Spring Festival](#)
Xiangyuan He, ..., and Weizheng Feng. EPJ Data Science, 2026. 0 citations.
Not measured Topic Match
No summary or abstract available
- [161] [SPARTAN: Data-Adaptive Symbolic Time-Series Approximation](#)
Fan Yang and John Paparizos. Proceedings of the ACM on Management of Data, 2025. 9 citations.
Not measured Topic Match
Abstract: Symbolic approximations are dimensionality reduction techniques that convert time series into sequences of discrete symbols, enhancing interpretability while reducing computational and storage costs. To construct symbolic representations, first numeric representations approximate and capture properties of raw time series, followed by a discretization step that converts these numeric dimensions into symbols. Despite decades of development, existing approaches have several key limitations that often result in unsatisfactory performance: they (i) rely on data-agnostic numeric approximations, disregarding intrinsic properties of the time series; (ii) decompose dimensions into equal-sized subspaces, assuming independence among dimensions; and (iii) allocate a uniform encoding budget for discretizing each...
- [162] [Representative Time Series Discovery for Data Exploration](#)
Ge Lee, ..., and Yanchang Zhao. Proc. VLDB Endow., 2024. 2 citations.
Not measured Topic Match
Abstract: In this work, we address the critical task of discovering representative time series in exploratory data mining. We define a representative time series, referred to as similarity-bounded representative time series, as one that represents other time series if their similarity meets a user-defined threshold. Building on this definition, we study the problem of finding the smallest set of such time series that can represent a specified proportion of all time series within the dataset. The representativeness of each similarity-bounded representative time series is controllable and determined by the specified level of similarity, and only the minimum number of such representatives...
- [163] [Efficient Semantic Trajectory Similarity Search](#)
Jian Chen, ..., and Jianzhong Li. IEEE Internet of Things Journal, 2025. 1 citations.
Not measured Topic Match
Abstract: Trajectory similarity search is a basic operation in spatial data analysis for nonterrestrial networks applications, such as asset tracking, maritime transportation, and urban air mobility. In many real-world applications (e.g., Flickr, Bikely, and Foursquare), users are allowed to share comments or experiences along with their trajectories, called semantic trajectories. In this article, we study a new type of similarity search on semantic trajectories, which takes into account spatial and textual information together. Given a query trajectory $\langle \text{LaTeX} \rangle_{t_{\text{q}}}$ and a similarity threshold $\langle \text{LaTeX} \rangle_{\theta}$, semantic trajectory similarity search retrieves all trajectories whose spatial-textual similarity to...
- [164] [An exhaustive comparison of distance measures in the classification of time series with 1NN method](#)
Tomasz Górecki, ..., and Paweł Piasecki. J. Comput. Sci., 2024. 13 citations.
Not measured Topic Match
No summary or abstract available
- [165] [Semantic Novelty at Scale: Narrative Shape Taxonomy and Readership Prediction in 28,606 Books](#)
W. Zimmerman. ArXiv, 2026. 0 citations.
Not measured Topic Match
Abstract: I introduce semantic novelty--cosine distance between each paragraph's sentence embedding and the running centroid of all preceding paragraphs--as an information-theoretic measure of narrative structure at corpus scale. Applying it to 28,606 books in PG19 (pre-1920 English literature), I compute paragraph-level novelty curves using 768-dimensional SBERT embeddings, then reduce each to a 16-segment Piecewise Aggregate Approximation (PAA). Ward-linkage clustering on PAA vectors reveals eight canonical narrative shape archetypes, from Steep Descent (rapid convergence) to Steep Ascent (escalating unpredictability). Volume--variance of the novelty trajectory--is the strongest length-independent predictor of readership (partial $\rho = 0.32$), followed by speed ($\rho = 0.19$) and Terminal/Initial...
- [166] [Time warping with Hellinger elasticity](#)
Y. Billig. ArXiv, 2026. 0 citations.
Not measured Topic Match
Abstract: We consider a matching problem for time series with values in an arbitrary metric space, with the stretching penalty given by the Hellinger kernel. To optimize this matching, we introduce the Elastic Time Warping algorithm with a cubic computational complexity.
- [167] [TSRF-Dist: a novel time series distance based on extremely randomized canonical interval forests](#)
A. Azzari, ..., and Pietro Sala. Data Mining and Knowledge Discovery, 2025. 0 citations.
Not measured Topic Match
Abstract: This paper presents TSRF-Dist, a novel distance between time series based on Random Forests (RFs). We extend to the time-series domain concepts and tools of RF distances, a recent class of robust data-dependent distances defined for vectorial representations, thus proposing the first RF distance for time series. The distance is determined by (i) creating an RF to model a set of time series, and (ii) exploiting the trained RF to quantify the similarity between time series. As for the first step, we introduce in this paper the Extremely Randomized Canonical Interval Forest (ERCIF), a novel extension of Canonical Interval Forests that...
- [168] [Correlation Based Dynamic Time Warping for ECG Waveform](#)
R. Lee, ..., and DaeEun Kim. Applied Sciences, 2026. 0 citations.
Not measured Topic Match
Abstract: Electrocardiogram waveform delineation is a fundamental task for quantitative cardiac analysis, yet accurate and consistent estimation of waveform boundaries remains challenging due to heart rate variability, inter-subject morphological differences, and nonlinear temporal distortions across cardiac cycles. Conventional rule-based methods and pointwise Dynamic Time Warping approaches are sensitive to amplitude variations and baseline fluctuations, while deep learning-based models require large annotated datasets and often suffer from limited interpretability and generalization. In this study, we propose a morphology-oriented ECG waveform alignment framework based on Pearson correlation-based Dynamic Time Warping (PCDTW). By integrating window-level matching with a correlation-driven cost function, the proposed method...
- [169] [An accurate slicing method for dynamic time warping algorithm and the segment-level early abandoning optimization](#)
Yuqi Luo, ..., and Sio-Kei Im. Knowl. Based Syst., 2024. 17 citations.
Not measured Topic Match
No summary or abstract available
- [170] [SimRN: Trajectory Similarity Learning in Road Networks based on Distributed Deep Reinforcement Learning](#)
Danlei Hu, ..., and Tianyi Li. Proc. VLDB Endow., 2025. 1 citations.
Not measured Topic Match
Abstract: Trajectory similarity computation in road networks is crucial for data analytics. However, both non-learning-based and learning-based methods face challenges. First, they suffer from low accuracy due to manual parameter selection for model training and the omission of key spatio-temporal features in road networks. Second, they have low efficiency, stemming from the high time complexity of similarity computation and the time-consuming training process. Third, learning-based methods struggle with poor model generality due to the small size of available training samples. To address these challenges, we propose an effective and efficient trajectory similarity learning framework for road networks, called SimRN. To our...
- [171] [TWHd: An Algorithm of Traveling Companion Discovery Based on Spatio-Temporal Trajectory](#)
Jiahuai Zhang, ..., and Shuo Li. 2025 11th International Conference on Computer and Communications (ICCC), 2025. 0 citations.
Not measured Topic Match

Abstract: The explosive growth in trajectory data has drawn the attention of academia to related analysis, among which traveling companion discovery is believed to be highly valuable in commercial applications and social studies. The key to improving discovery accuracy is to measure the similarity between two trajectories more precisely. Although most of the existing trajectory similarity methods are time-insensitive, certain timesensitive methods align two trajectories according to the time index of the trajectory points. In this paper, we propose a novel metric, Time-Warping Hausdorff Distance(TWHD), to qualify the spatial-temporal trajectory similarity. The TWHD(Time-Warping Hausdorff Distance) not only utilizes the time index...

[172] [HivePool: An Exploratory Visualization System for Honey Beehive Data](#)

Tinghao Feng, ..., and R. Tashakkori. Journal of Engineering Research and Sciences, 2024. 0 citations.

Not measured Topic Match

Abstract: : Honey bee health is crucial for global ecosystems, but traditional data analysis methods often struggle to capture the complex interplay between bee behavior and environmental factors. To bridge this gap, we developed HivePool, a novel data visualization and analysis tool designed to empower beekeepers and researchers with deeper insights into these interactions. This paper explores HivePool's functionalities, focusing on its interactive visualizations and innovative time-oriented pattern recognition for event prediction. By leveraging time series visualization techniques, HivePool allows users to explore not only static relationships between environmental variables but also how these variables change dynamically leading up to specific...

[173] [A New Clustering Strategy for Geo Referenced Time Series Based on Optimal Transport](#)

Pasquale Pipiciello, ..., and Gianmarco Borrata. Environmetrics, 2026. 0 citations.

Not measured Topic Match

Abstract: Advances in spatio temporal data collection have created a demand for efficient methods to analyze geo referenced time series (GTS), which capture changes over time at specific spatial locations. Traditional clustering methods often struggle to handle the high dimensional, complex nature of GTS. This paper proposes a novel approach for clustering GTS using a new dissimilarity measure, the Bi Gromov Dynamic Time Warping (Bi GDTW) distance. This method combines the alignment based Dynamic Time Warping framework with Gromov-Wasserstein distance to account for both temporal and spatial dependencies in the data. The proposed measure supports clustering through algorithms such as K means, enabling effective pattern discovery in GTS...